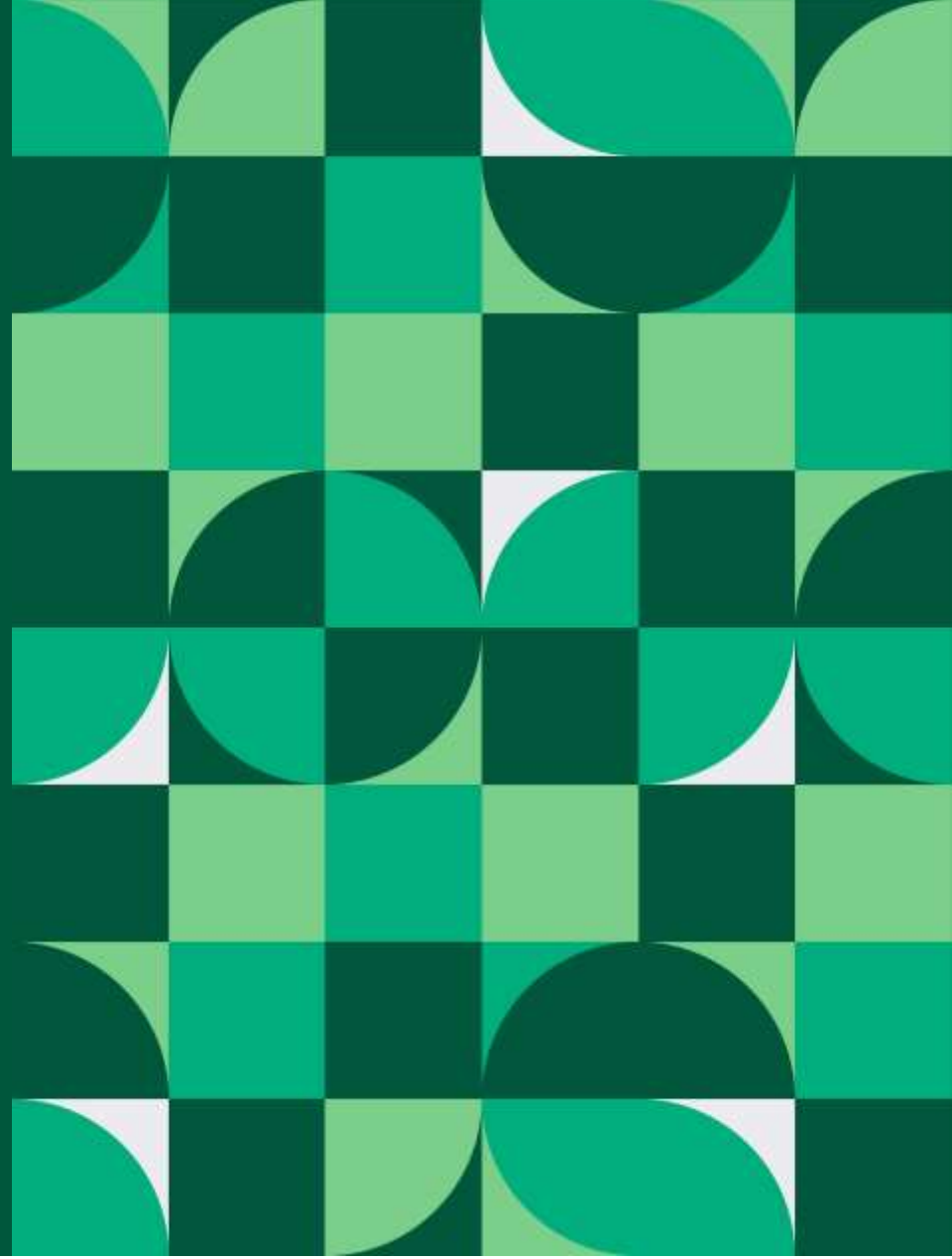
No Coding Required!!

Real Time Test & Control Applications



EMERSON



Meet Dan

20	2	22
Years at NI	Years in Industry	Years LabVIEW

Background & Skills

- Distinguished FAE; Certified LabVIEW Architect; Certified LabVIEW Embedded Developer
- Software Dev, Systems Management, HiL, Protocols, CompactRIO, & FPGAs
- Previous life automating automotive test cells

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Agenda

DIY Software	Challenge: Handling Software at Scale
What is NI VeriStand?	The Stress Test
The Stress Test…	Demo: NI VeriStand as Embedded Logger
	Demo: NI VeriStand as Control System
	Demo: NI VeriStand as Integrated Test System
Conclusion	Wrap Up

DIY Software Pros & Cons

Pro: Self Reliant

Software can directly match requirements

No one limiting you on bug fixes, new features, etc

Code is owned by you

Original developers know how everything works

Con: Scale

All the pros become cons at two unique apps…

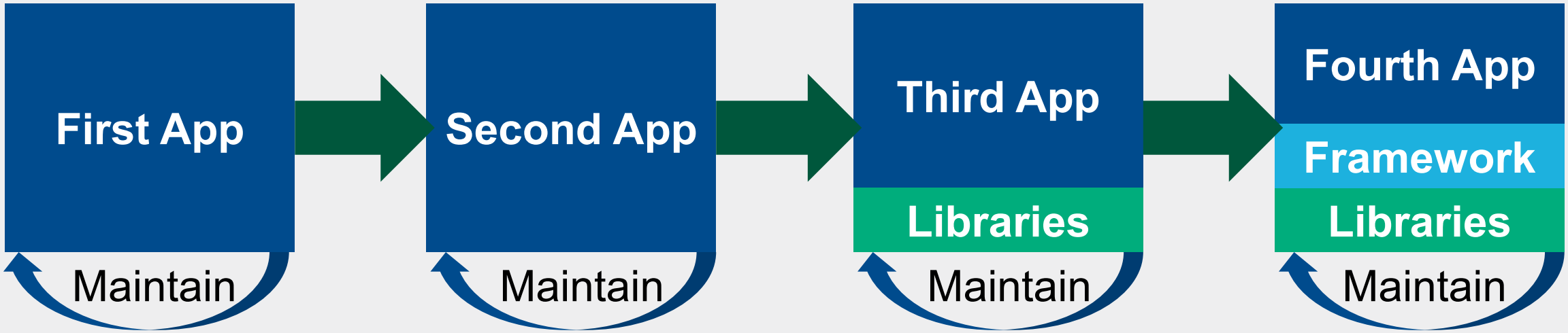
2x the coding (lost time)

2x the maintenance / upgrades (lost time)

2x the learning curve (lost time)

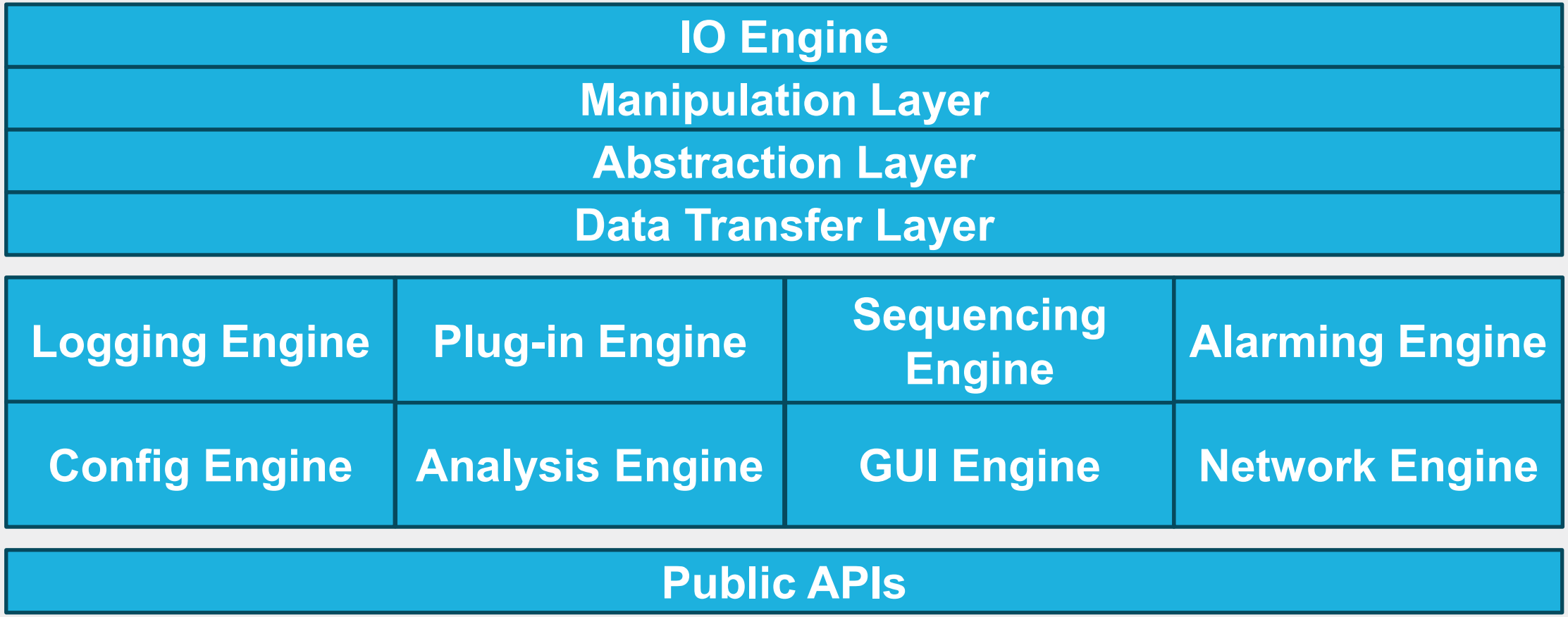
Requires a special set of skills (resource constraint)

The Natural Progression Towards Standardization…



Common Core Framework Pieces

** Someone must maintain the framework



EMERSON

Enter NI VerStand

** NI owns the maintenance



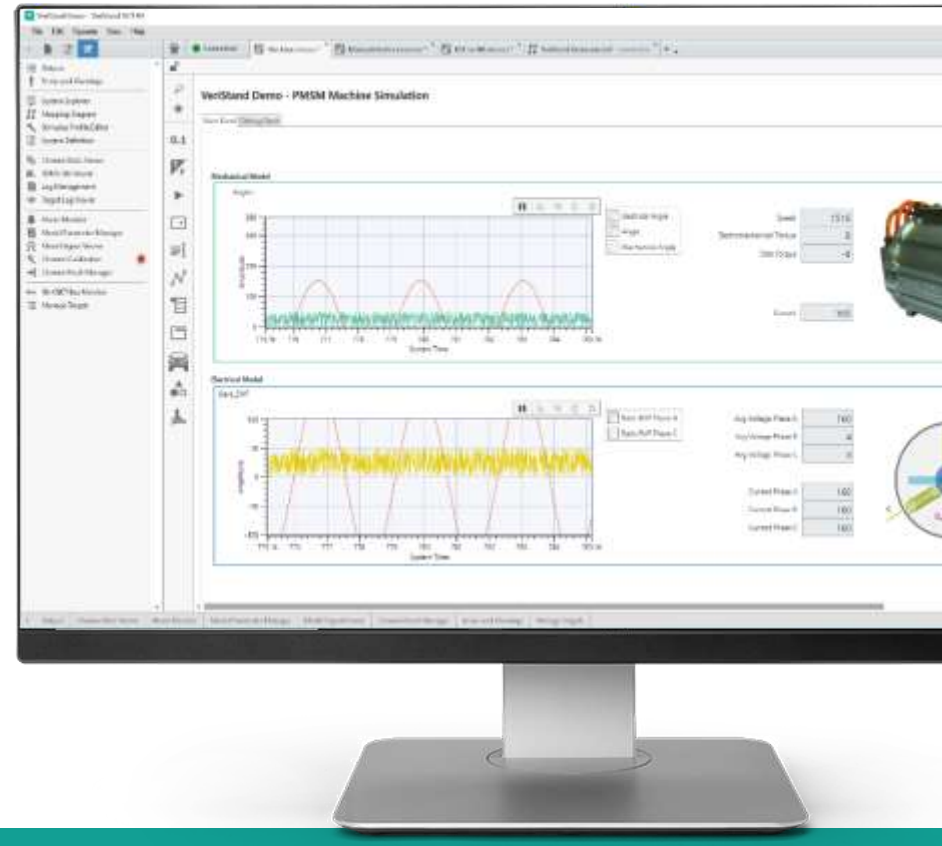
VeriStand™

IO Engine			
Manipulation Layer			
Abstraction Layer			
Data Transfer Layer			
Logging Engine	Plug-in Engine	Sequencing Engine	Alarming Engine
Config Engine	Analysis Engine	GUI Engine	Network Engine
Public APIs			

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VeriStand™

Validates hardware and performs embedded software test for HIL with model integration, real-time stimulus generation, and an extensible software environment.



Ready to Run Software for HIL & Real Time Test
Create, deploy, and leverage closed loop control on Linux RT

Quickly Configure Systems
Configure I/O channels, alarming, data logging, stimulus generation, fault insertion, user interfaces and bus communications

Model Integration
Integrate plant and controller models from Simulink and other 3rd party modeling environments adhering to the FMI standard

Customizability
Adapt to changing requirements with an extensible software environment.
Build plug-ins for 3rd party hardware and additional measurement types

Test Automation
Use test sequences and scripting to configure, orchestrate systems through TestStand, .NET APIs, Python, and ASAM XIL and integrate with CI/CD workflows

Connect VeriStand

With **LabVIEW** to develop plugins to extend your test system with additional functionality

With **TestStand** to build libraries of test steps for higher level orchestration

With **your test infrastructure and IP** to maximize reuse

EMERSON

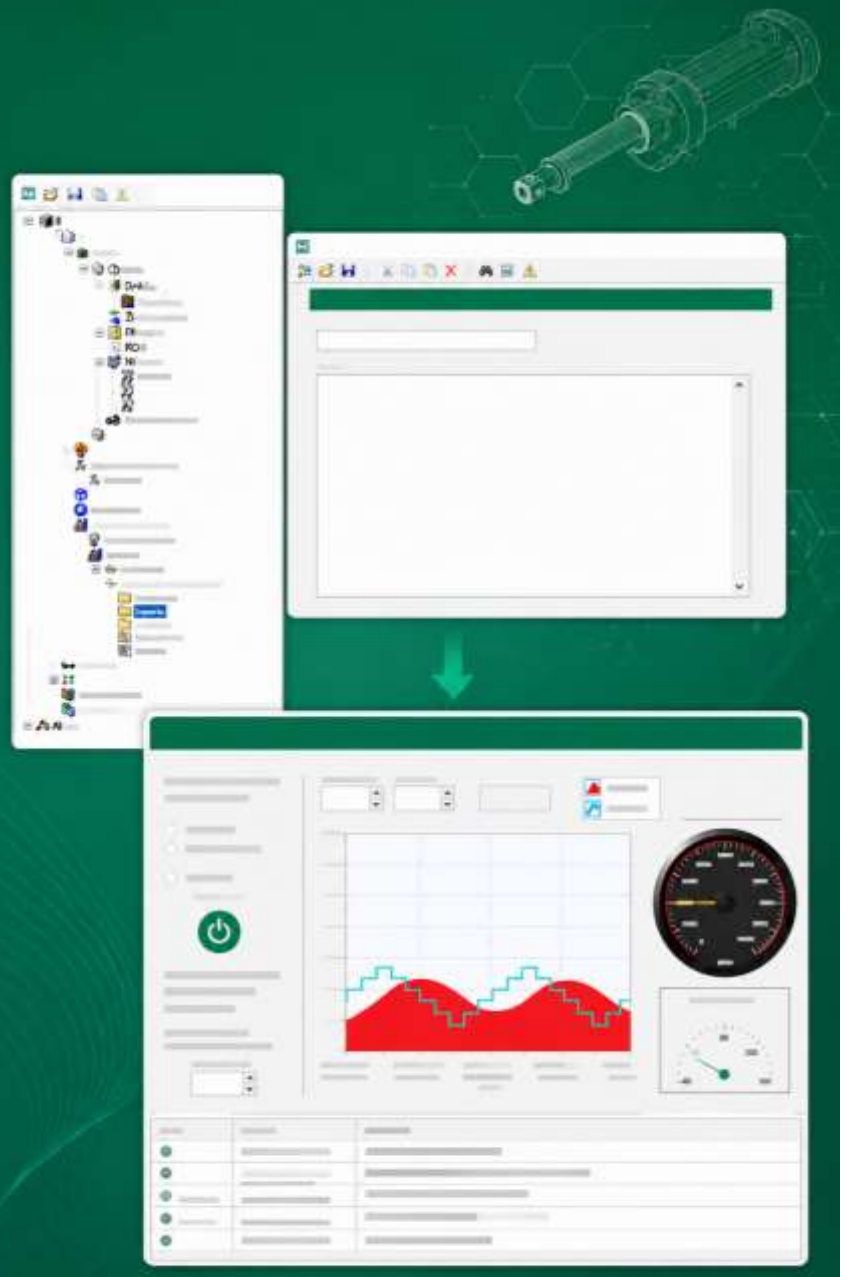
Deployable Framework to NI Linux Real Time Targets

VeriStand™

IO Engine			
Manipulation Layer			
Abstraction Layer			
Data Transfer Layer			
Logging Engine	Plug-in Engine	Sequencing Engine	Alarming Engine
Config Engine	Analysis Engine	GUI Engine	Network Engine
Public APIs			

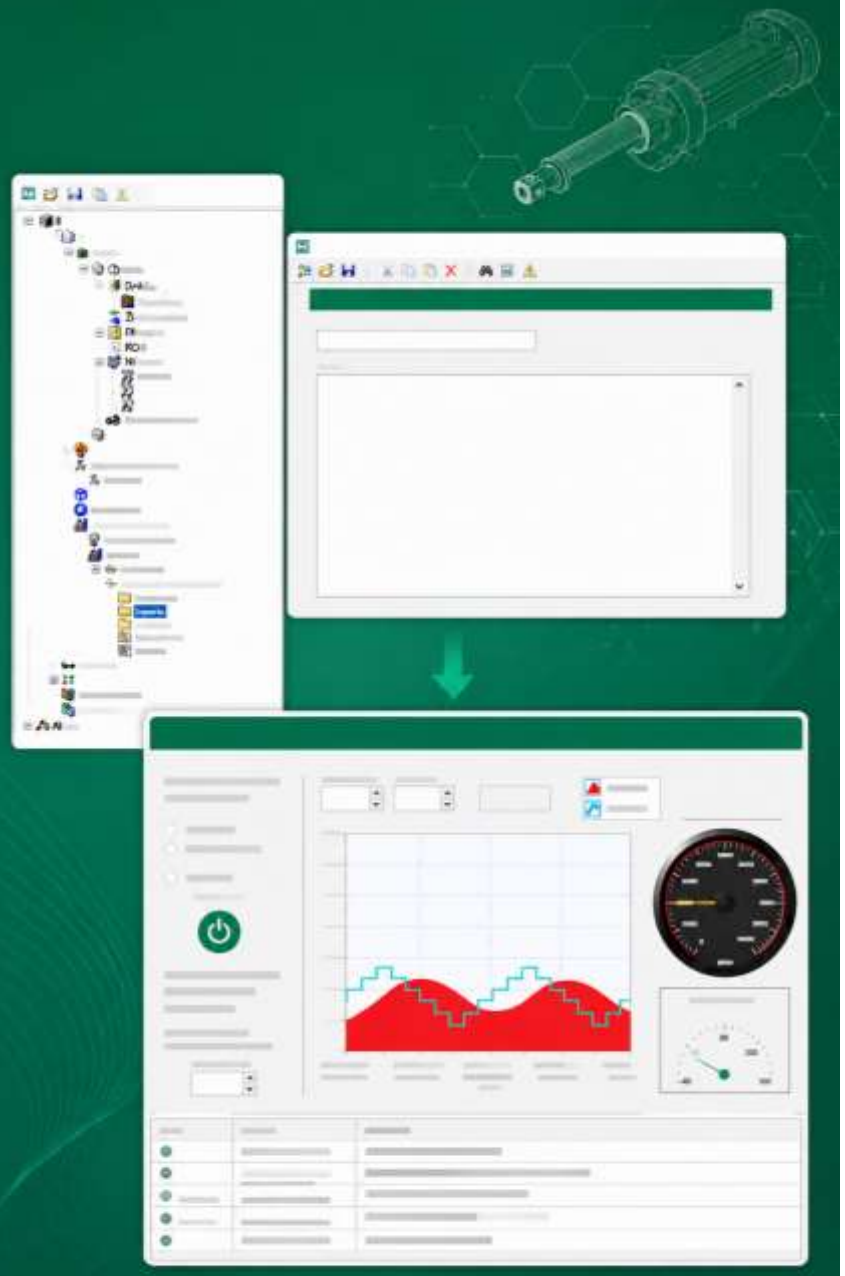






Features

Hardware	Configure I/O and smart bus hardware for real time execution
Simulation	Import and exercise models from simulation environments (Simulink, LabVIEW, etc.)
Manipulation	Manipulate data through calibration, scales, alarms, mapping, and logging
Extensions	Add custom features, 3rd party hardware, and specialized UI tools directly in the application
UI	Configure multi-page user interfaces and use application-specific tools
Automation	Create deterministic sequences that execute in real time
Interoperability	API for everything… Integrate and customize with LabVIEW, Python, TestStand, etc.



Features

Hardware	Configure I/O and smart bus hardware for real time execution	System Definition
Simulation	Import and exercise models from simulation environments (Simulink, LabVIEW, etc.)	System Definition
Manipulation	Manipulate data through calibration, scales, alarms, mapping, and logging	System Definition
Extensions	Add custom features, 3rd party hardware, and specialized UI tools directly in the application	Custom Devices
UI	Configure multi-page user interfaces and use application-specific tools	Workspace
Automation	Create deterministic sequences that execute in real time	Profile Editor / APIs
Interoperability	API for everything… Integrate and customize with LabVIEW, Python, TestStand, etc.	API

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NI VeriStand Applications

Best Fit

Hardware in the Loop

Test cell applications

Control applications

FPGA applications

Simulation applications

Embedded logging

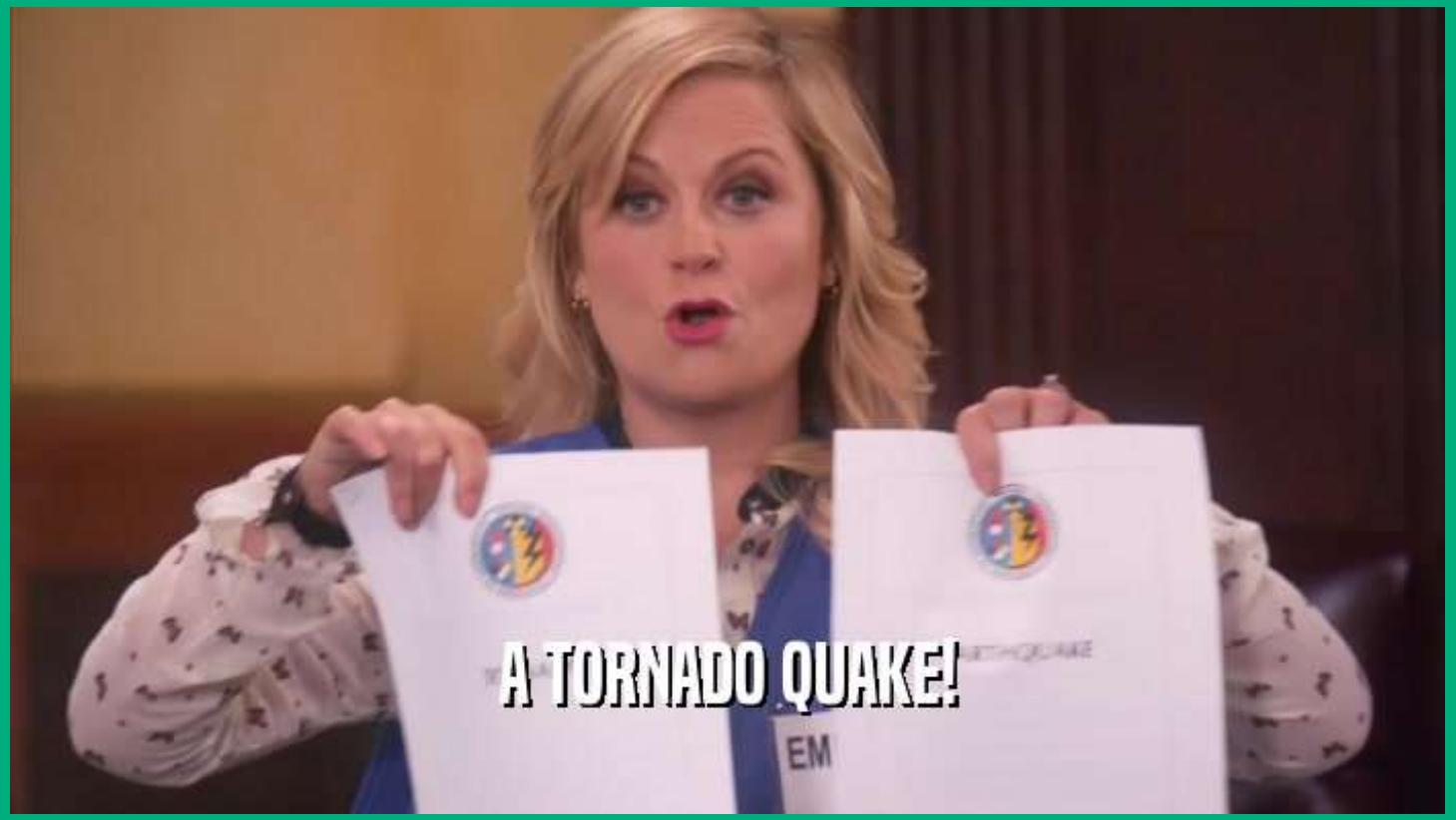
Deterministic or simple sequencing applications

Waveform logging

Avoid

Waveform processing

Production test applications



The Stress Test...

Your boss walks in and says…

I need you to build something that:

- Is configurable
- Is deterministic & embedded
- Talks to hardware (NI & 3rd Party)
- Logs data
- Provides a GUI
- Controls a system
- Sequences through multiple setpoints
- Handles safeties

And… yeah, we need it now…



EMERSON

Now I need to just build…

A real-time execution engine

A hardware abstraction layer

Instrument drivers

Integrated control and safety

Sequencing engine

Logging infrastructure

A UI

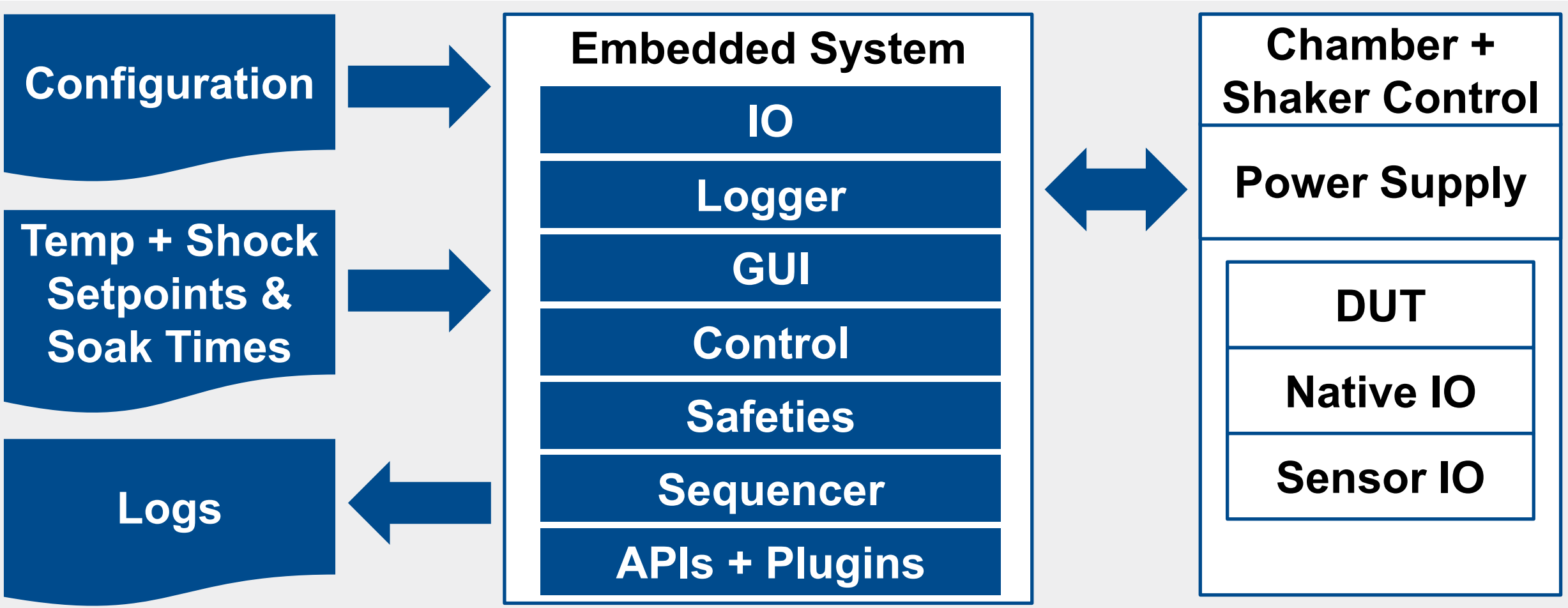
App packaging & deployment

Documentation

…



Demo Requirements



You Have One Hour

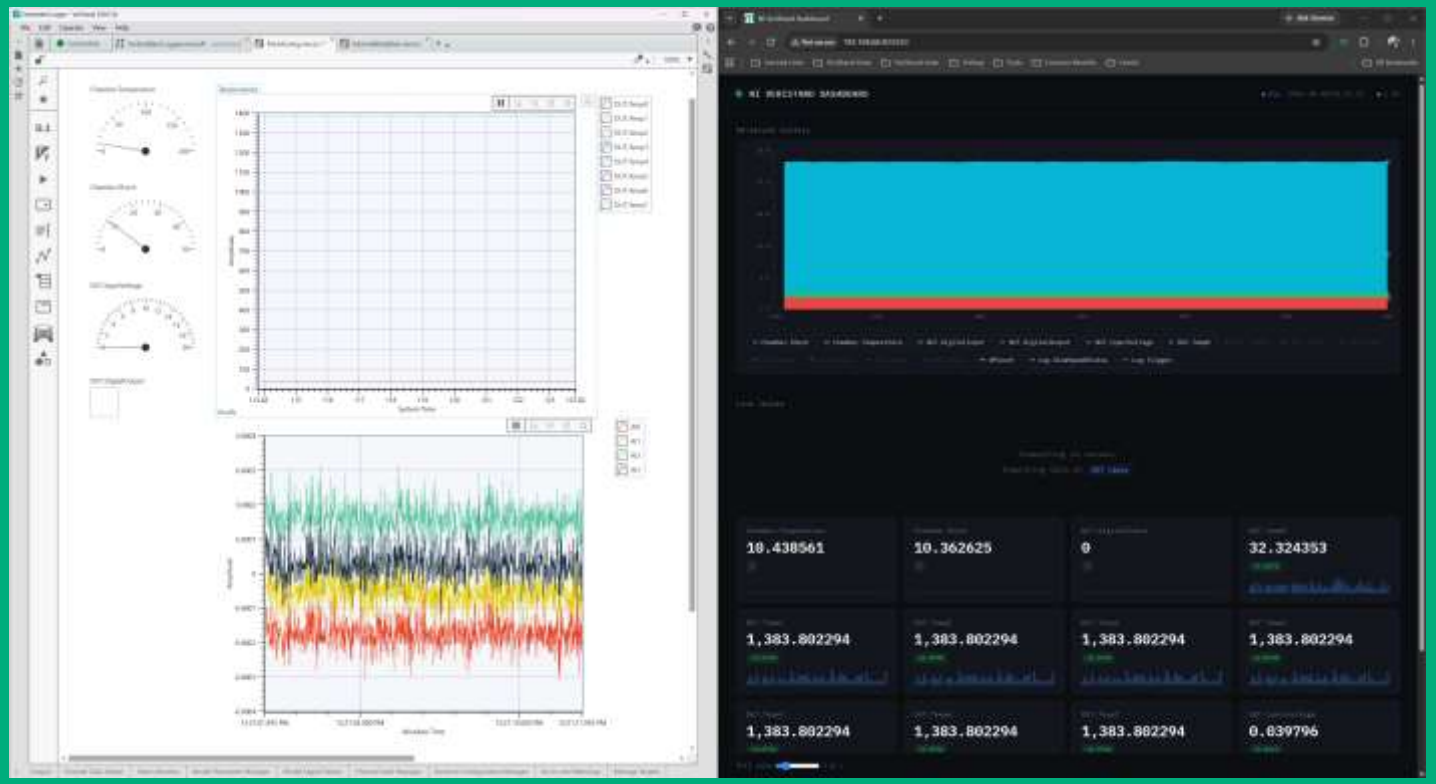
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Demo Requirements

- Is configurable
- Is deterministic & embedded
- Talks to NI hardware
- Provides a GUI
- Logs data
- Controls a system
- Handles safeties
- Sequences through multiple setpoints
- Talks to 3rd Party Hardware

- VeriStand System Explorer (Out of Box)
- VeriStand RT Engine (Out of Box)
- IO Addon (Out of Box)
- GUI + Web Addon (Out of Box)
- Embedded Data Logger (Out of Box)
- Control Addon (Out of Box)
- Alarms (Out of Box)
- RT Sequence (Out of Box)
- Instrument Addon (Out of Box)

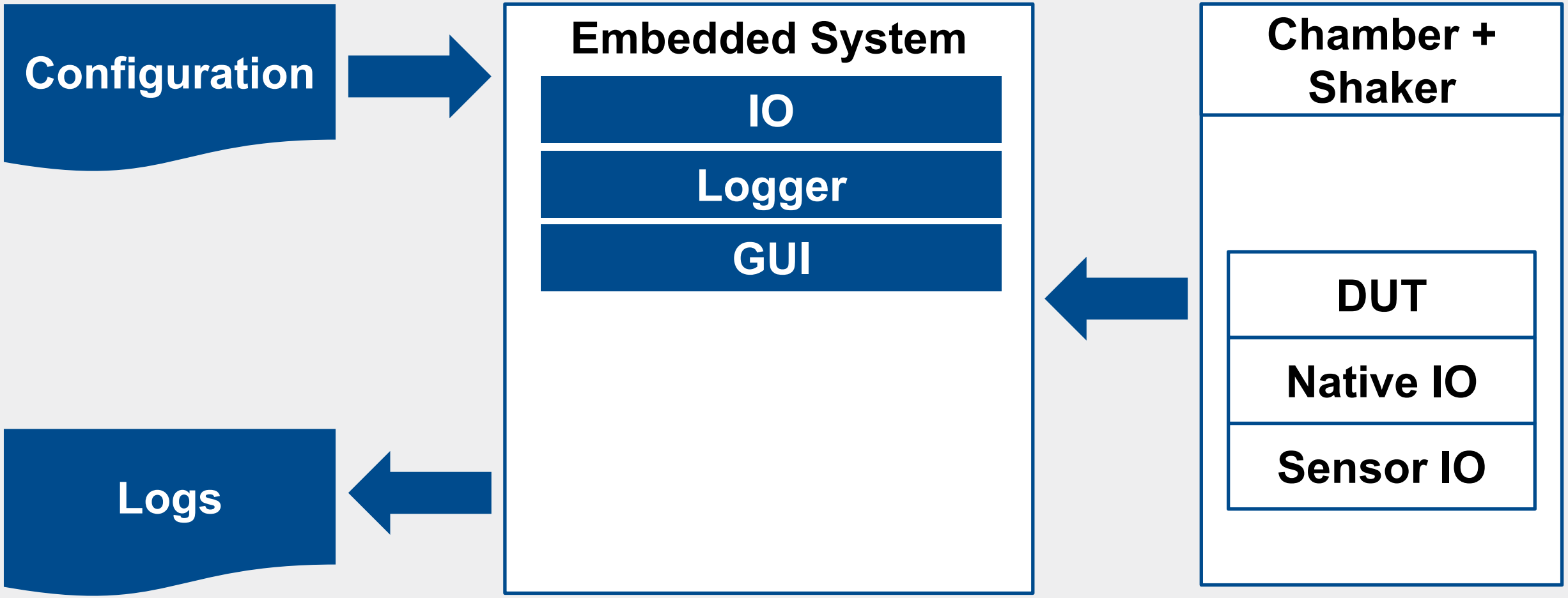
- **Just Configure and Go!**



Demo: NI VeriStand as Embedded Logger

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Demo: NI VeriStand as Embedded Logger



Demo: NI VeriStand as Embedded Logger

NI Input Channel List

DUT Instrumentation:

- 8 Thermocouples → NI 9213 (Mod1)
- 4 Accelerometers → NI 9234 (Mod3)

DUT IO

- Supply Voltage → NI 9221 (Mod2; ai0)
- 1 Digital Output → NI 9401 (Mod4; di0)

Chamber

- Temperature (C → Volt) → NI 9215 (Mod2; ai0)
- Shock (gs → Volt) → NI 9215 (Mod2; ai1)

NI VeriStand Features – Hardware Support

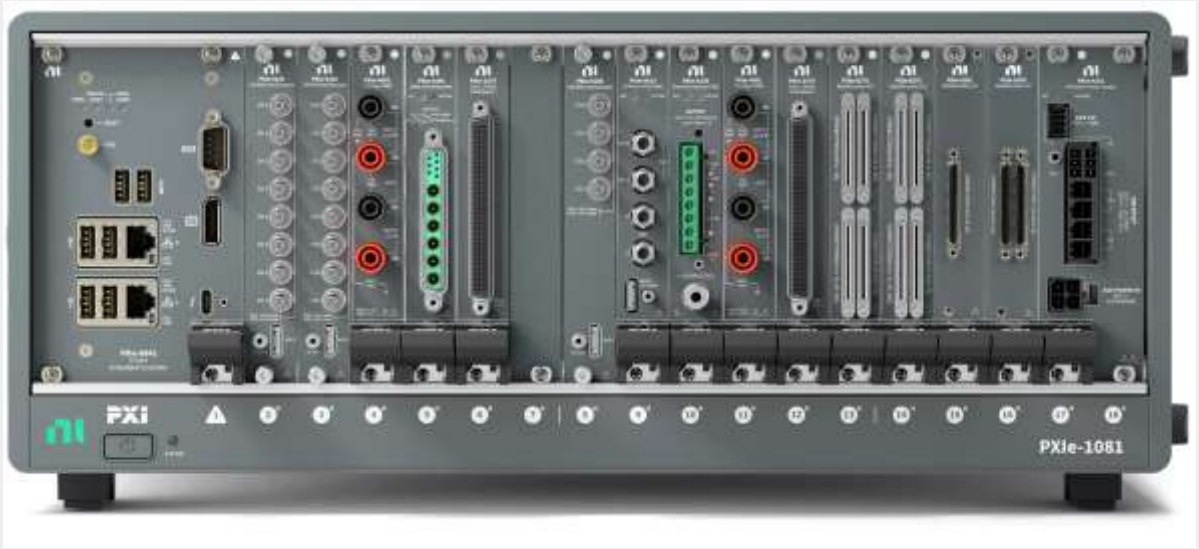
PXI (Windows & RT)

cRIO (RT) + EtherCAT Chassis

Most NI hardware types:

- SLSC
- FPGA
- DAQ
- XNET
- Timing and Syncing devices

Customize Support through Addons: https://github.com/NIVeriStandAdd-Ons



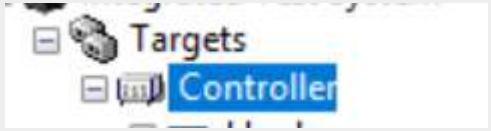


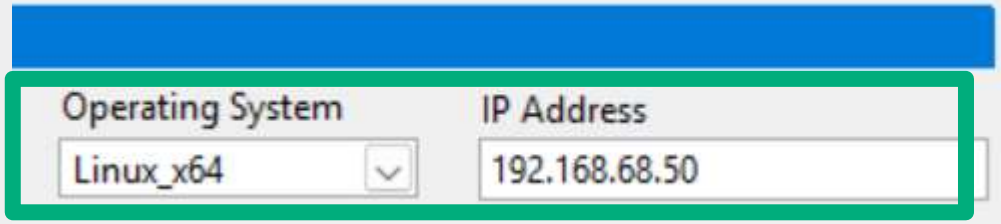
Configure Hardware – Embedded System

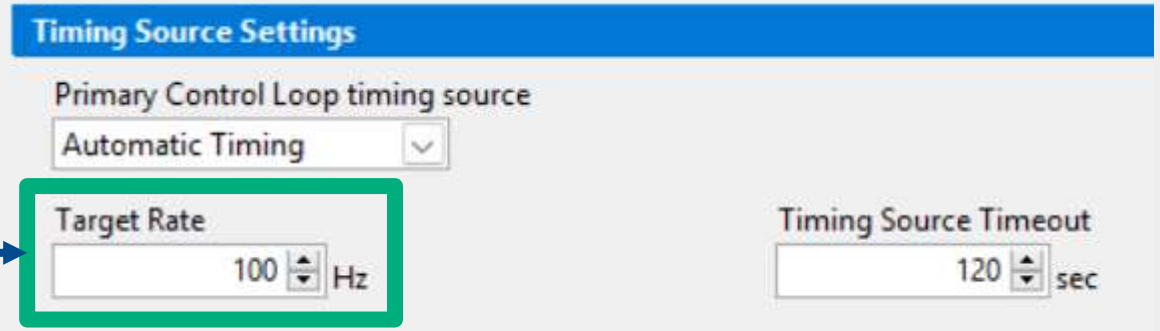
PXI Controller Settings

On Controller

- Set Operating System
- Set IP Address
- Set Target Rate (Loop rate)







Sets loop rate for most everything
(models, embedded logging, control, etc)

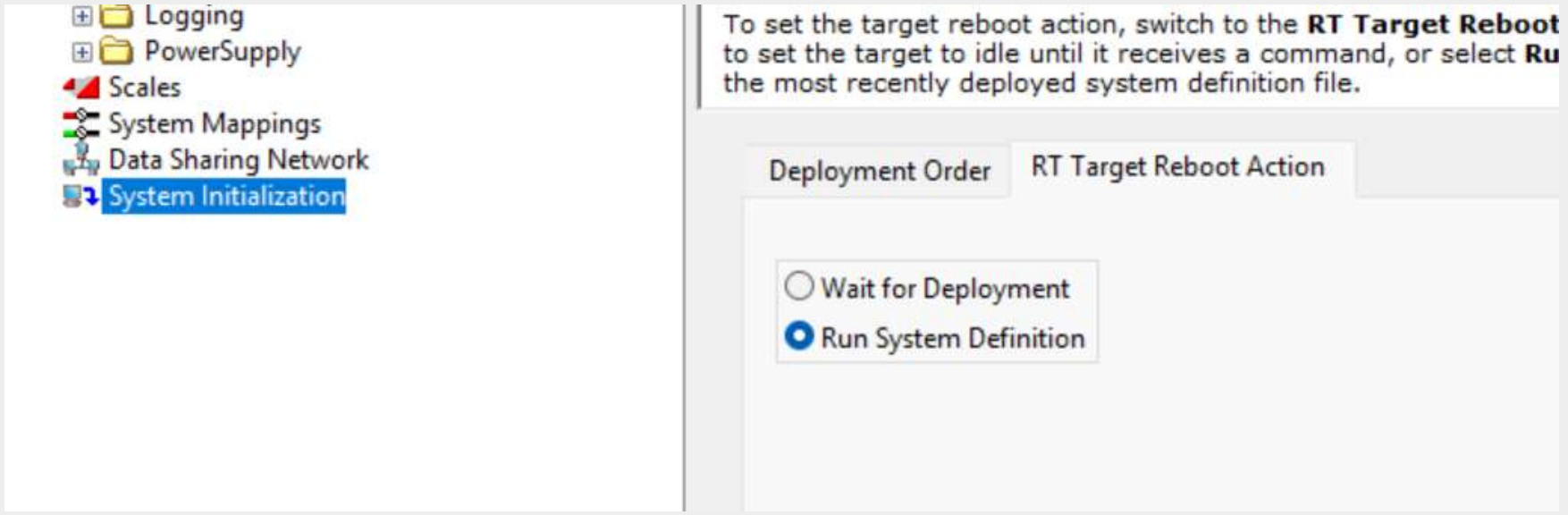
Configure Hardware – Embedded System

System Initialization

On System Initialization

On RT Target Reboot Action tab:

Run System Definition

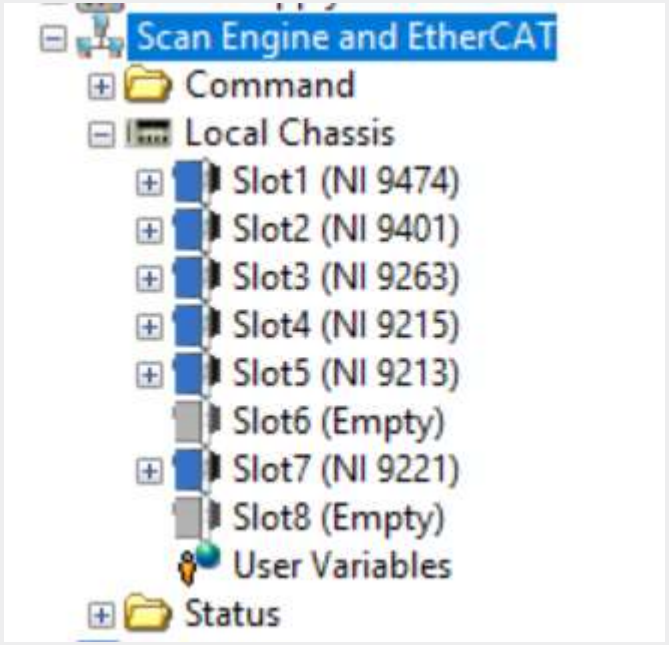


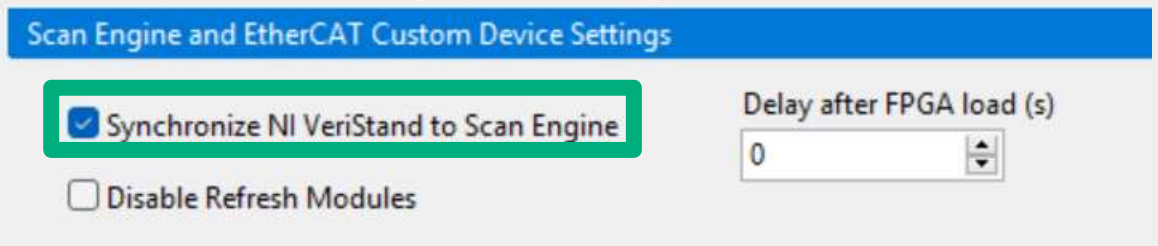
Configure Hardware – NI IO

Scan Engine Addon

On Scan Engine and EtherCAT

- Set Synchronize NI VeriStand to Scan Engine to True
- (keeps model, logging, control, etc synced)





Configure Hardware – NI IO

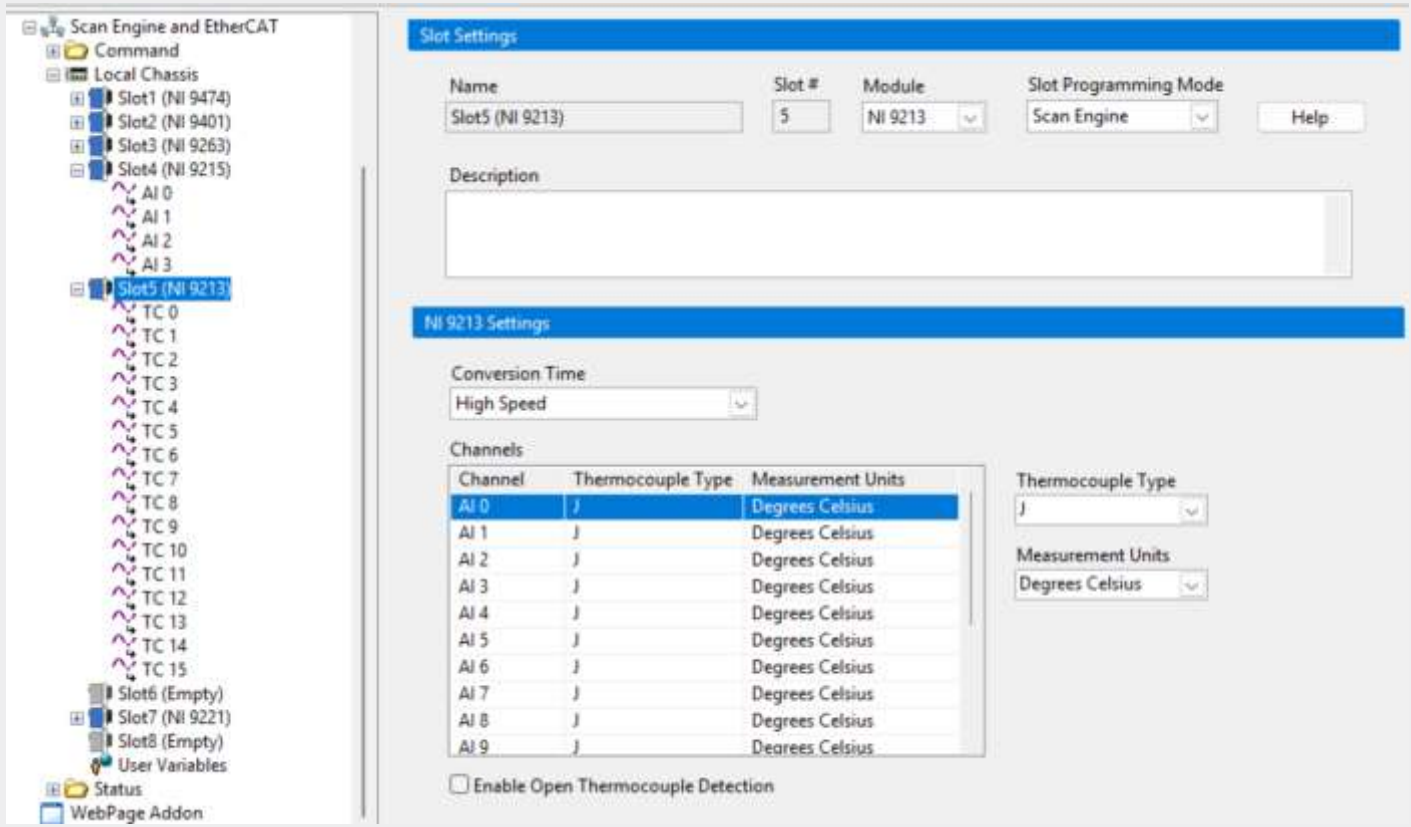
Scan Engine Addon

Add Local Chassis

Add Modules that are in physical system

Per Module

- Configure Programming Mode
- Configure IO Settings



Configure Hardware – NI IO

Alias Everything!!

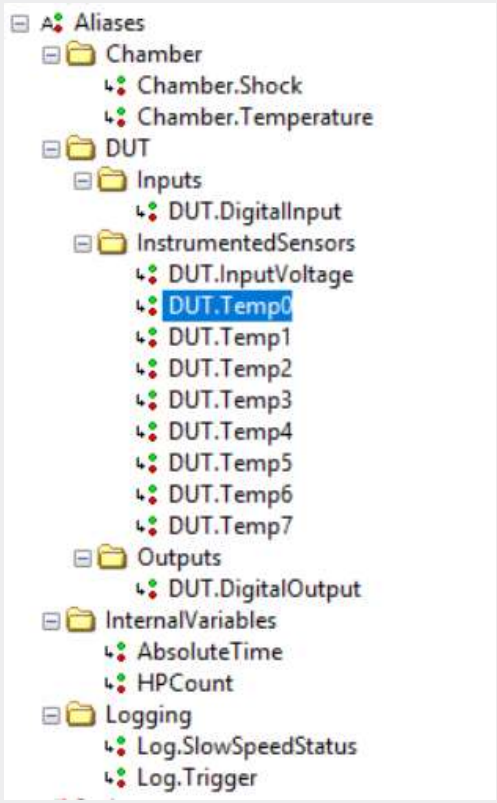
Standard, Abstracted Name

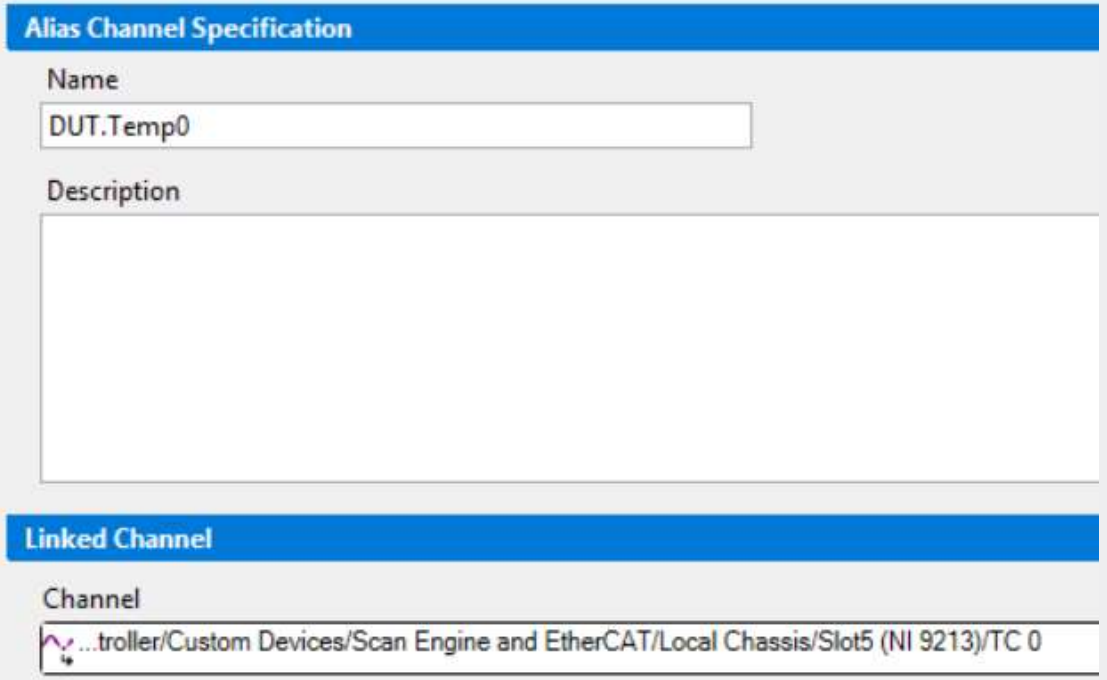
Points at physical channels

Link everything possible to the alias

Makes changes easier

Makes re-use easier

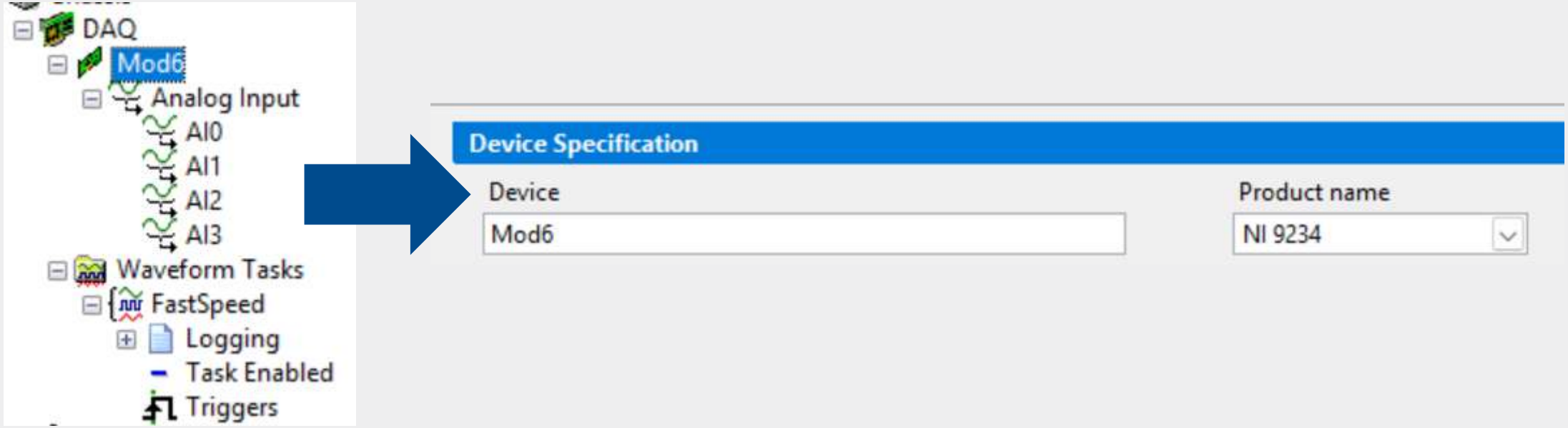




Configure Hardware – NI IO

NI DAQmx Waveforms

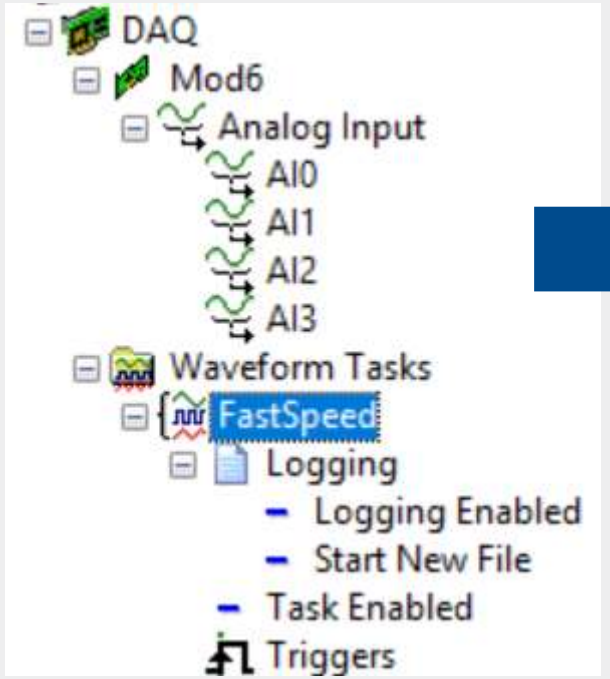
Add Modules that are in physical system

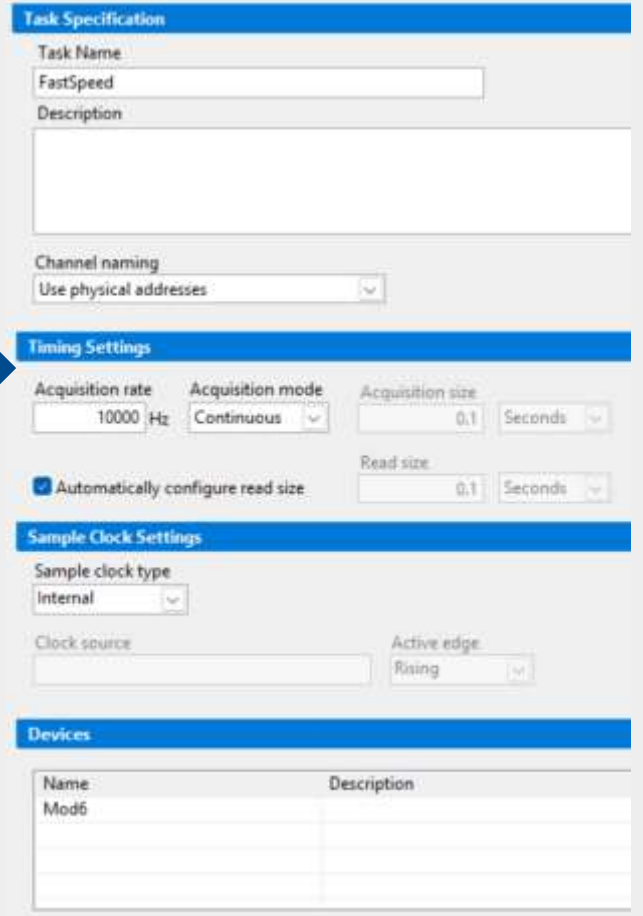


Configure Hardware – NI IO

NI DAQmx Waveforms

Create Task



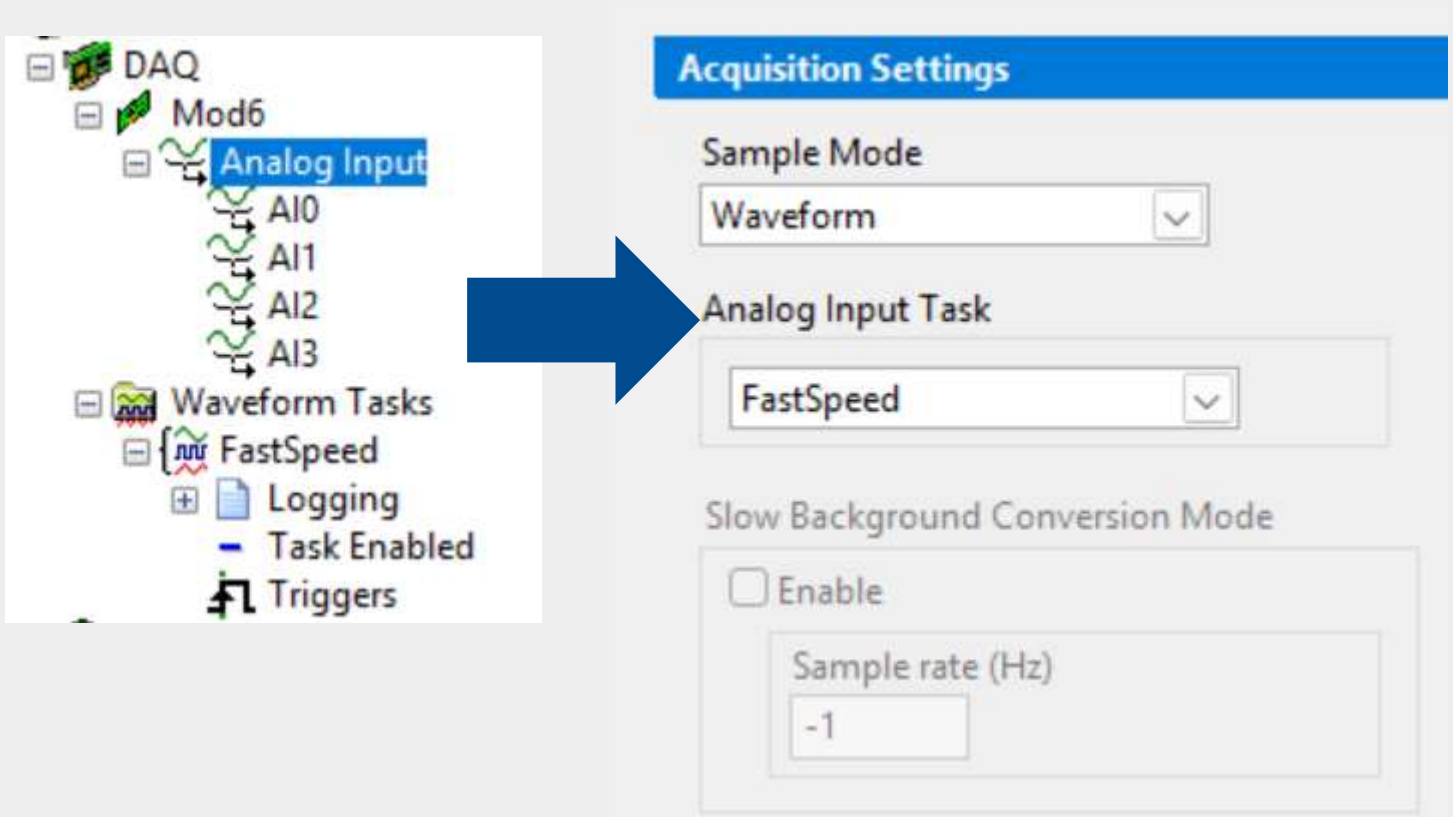


Configure Hardware – NI IO

NI DAQmx Waveforms

Set Sample Mode to Waveform

Assign Analog Input Task to previously created



NI VeriStand Features – Logging

Embedded Logger (On Target)

Host Side Logger (On Host)

Waveform Task Logging (On Target)

Base Path: Logs\

File Name: {Specification Name}\{Year}\{Month}\{Day}\Log File_{Year}_{Month}_{Day}_{Hour}_{Minute}_{Second}.tdms

GROUP	CHANNEL NAME	PATH
Data	ActualRPM	Aliases/ActualRPM
Data	DesiredRPM	Aliases/DesiredRPM
Data	EnginePower	Aliases/EnginePower
Data	EngineTemp	Aliases/EngineTemp
Diagnosti...	Actual Loop Rate	Targets/Controller/System Channels/Actual Loop Rate
Diagnosti...	Delta T	Targets/Controller/System Channels/Delta T
Diagnosti...	HP Loop Duration	Targets/Controller/System Channels/HP Loop Duration
Diagnosti...	HP-LP Overwrite	Targets/Controller/System Channels/HP-LP Overwrite
Diagnosti...	LP Count	Targets/Controller/System Channels/LP Count
Diagnosti...	LP Data Count	Targets/Controller/System Channels/LP Data Count
Diagnosti...	LP Loop Duration	Targets/Controller/System Channels/LP Loop Duration
Diagnosti...	Model Count	Targets/Controller/System Channels/Model Count
Diagnosti...	System Time - Microsecon...	Targets/Controller/System Channels/System Time - Microsecon...

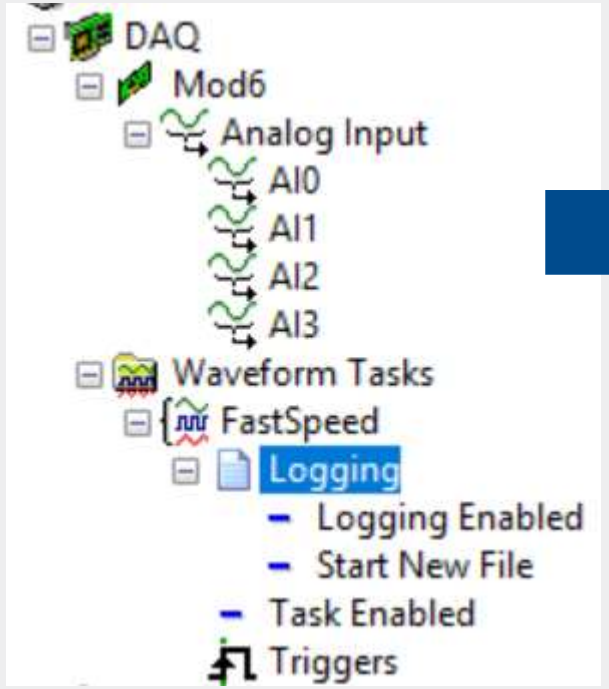
EMERSON

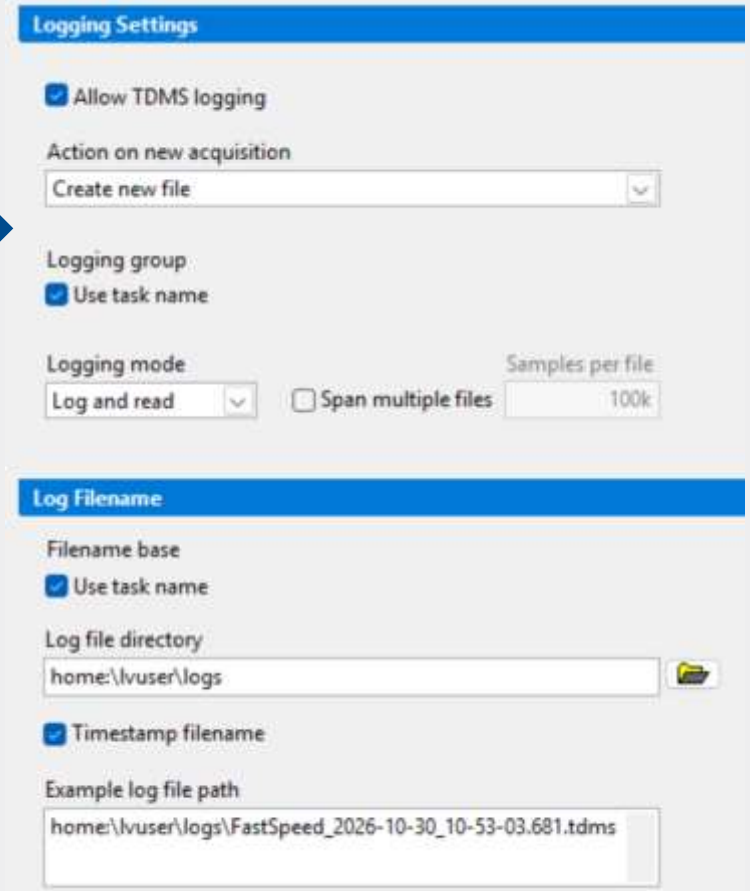
Configure Logging – Waveforms

Waveform Task Logging

On Logging

- Allow TDMS Logging = True
- Set Log Directory: home:\lvuser\logs





Configure Logging – Single Point Data

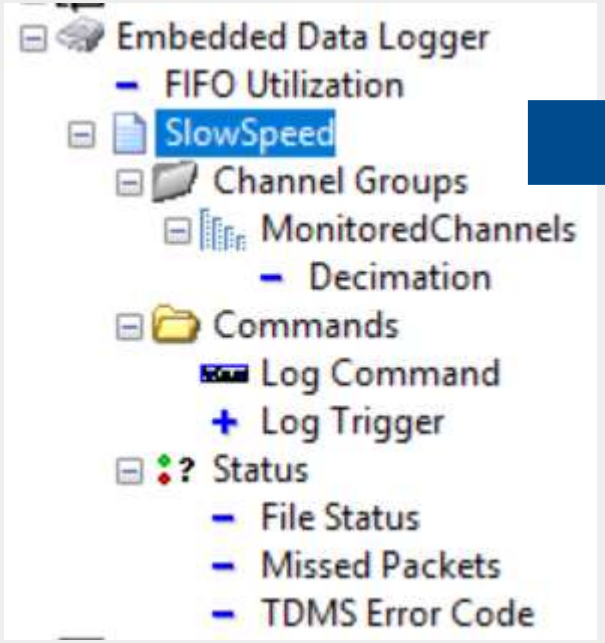
Embedded Data Logger

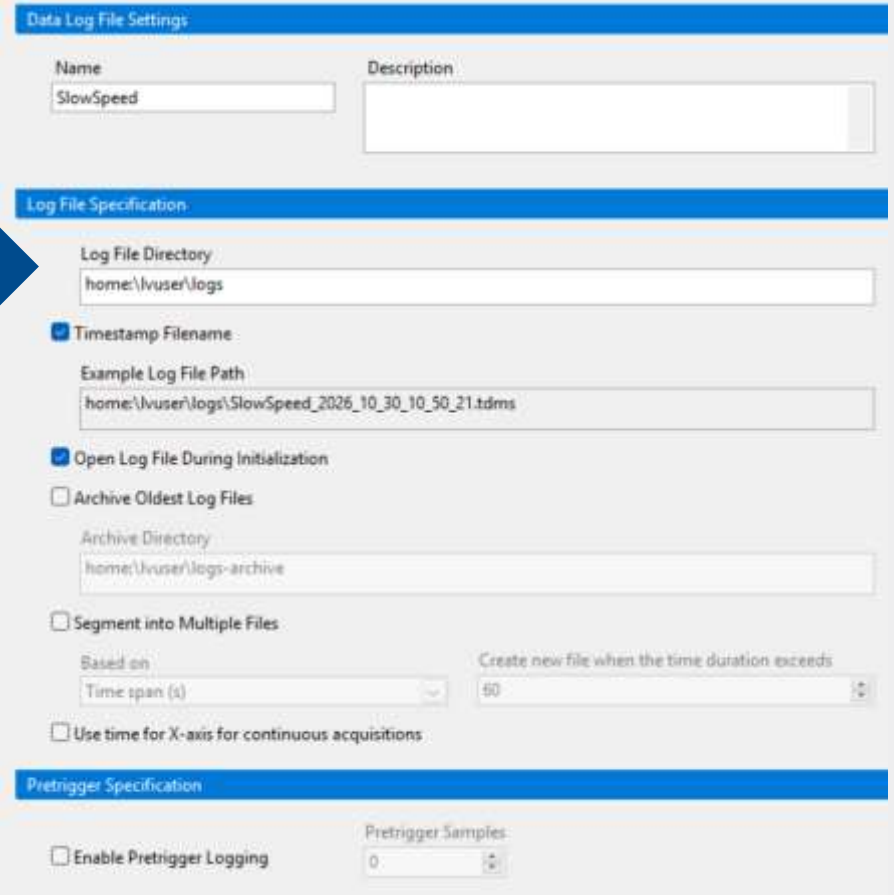
On Embedded Data Logger:

- Add Log File

On Log File:

- Set Log File Directory to home:\lvuser\logs
- Add Channel Group



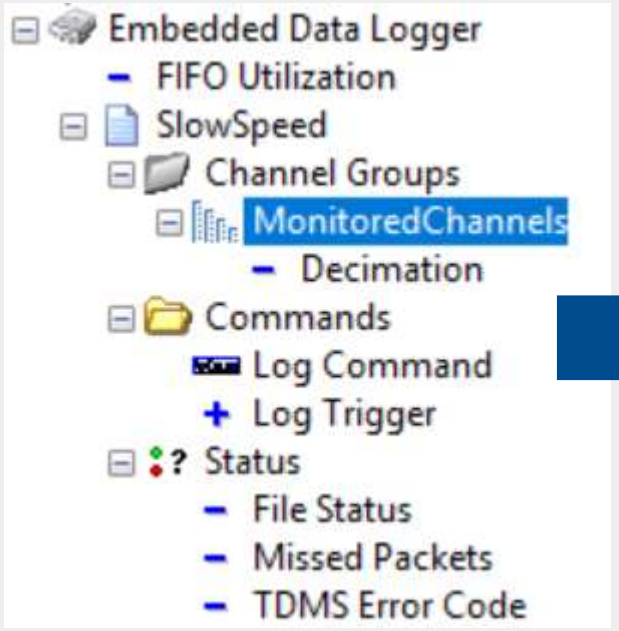


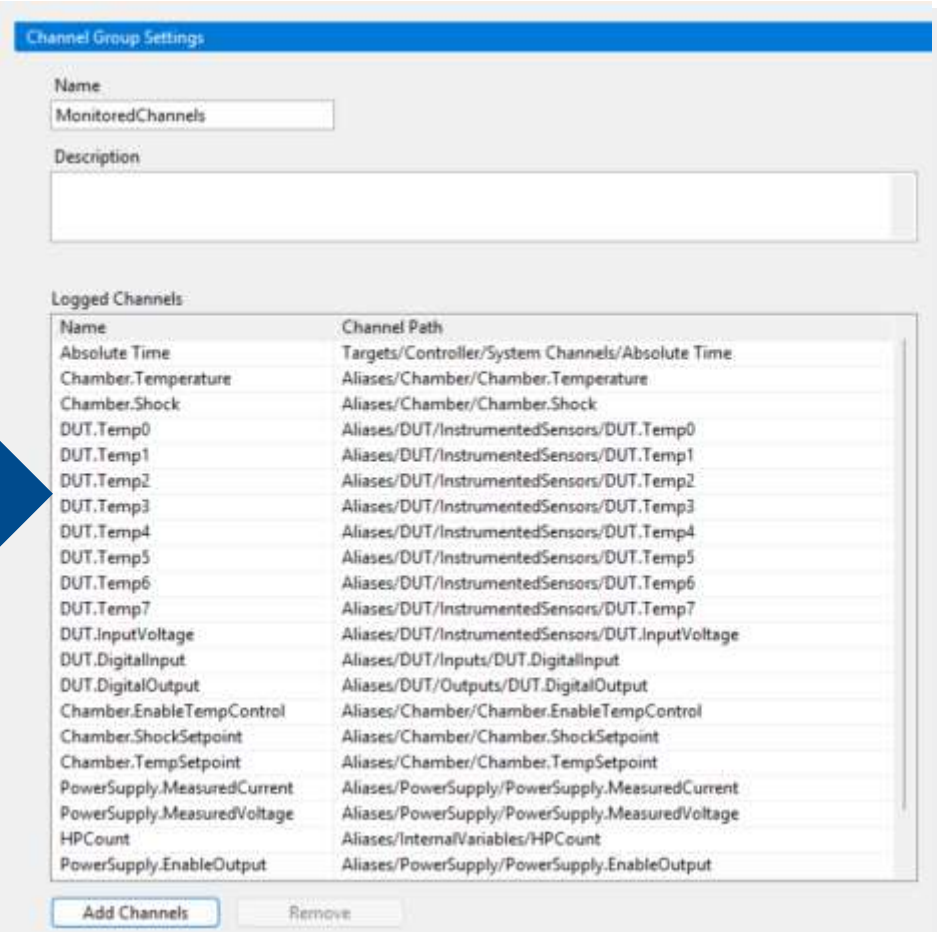
Configure Logging – Single Point Data

Embedded Data Logger

On the created Channel Group:

- Add Channels





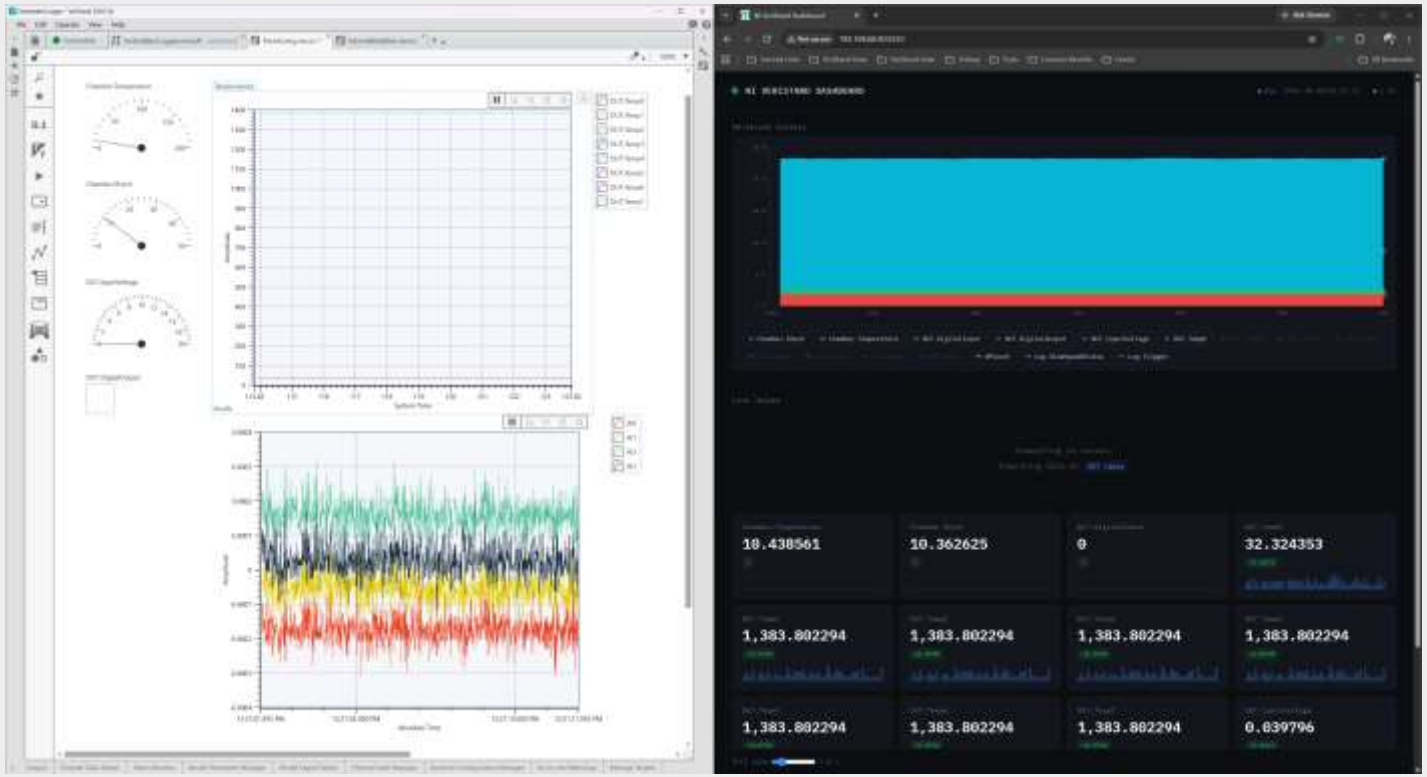
NI VeriStand Features – User Interface

User Configurable GUI

Multiple page support

Multiple tools for additional information

Customize Support through Tools, Addons, & APIs



Configure GUI – Hosted Web Page

WebPage Addon

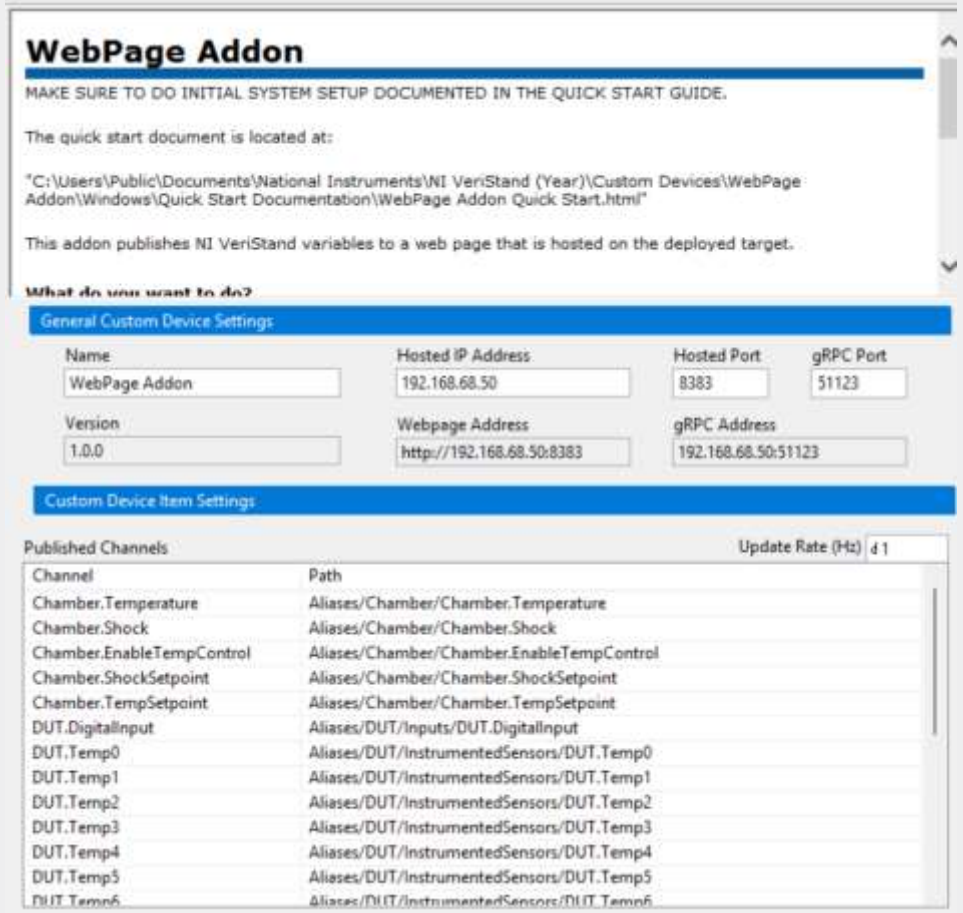
On Webpage Addon:

Select Channels

Set IP Address and Port

Source Code:

https://github.com/NIVeriStandAdd-Ons/NI-Veristand-Webpage-Addon



Configure GUI – Host Side UI

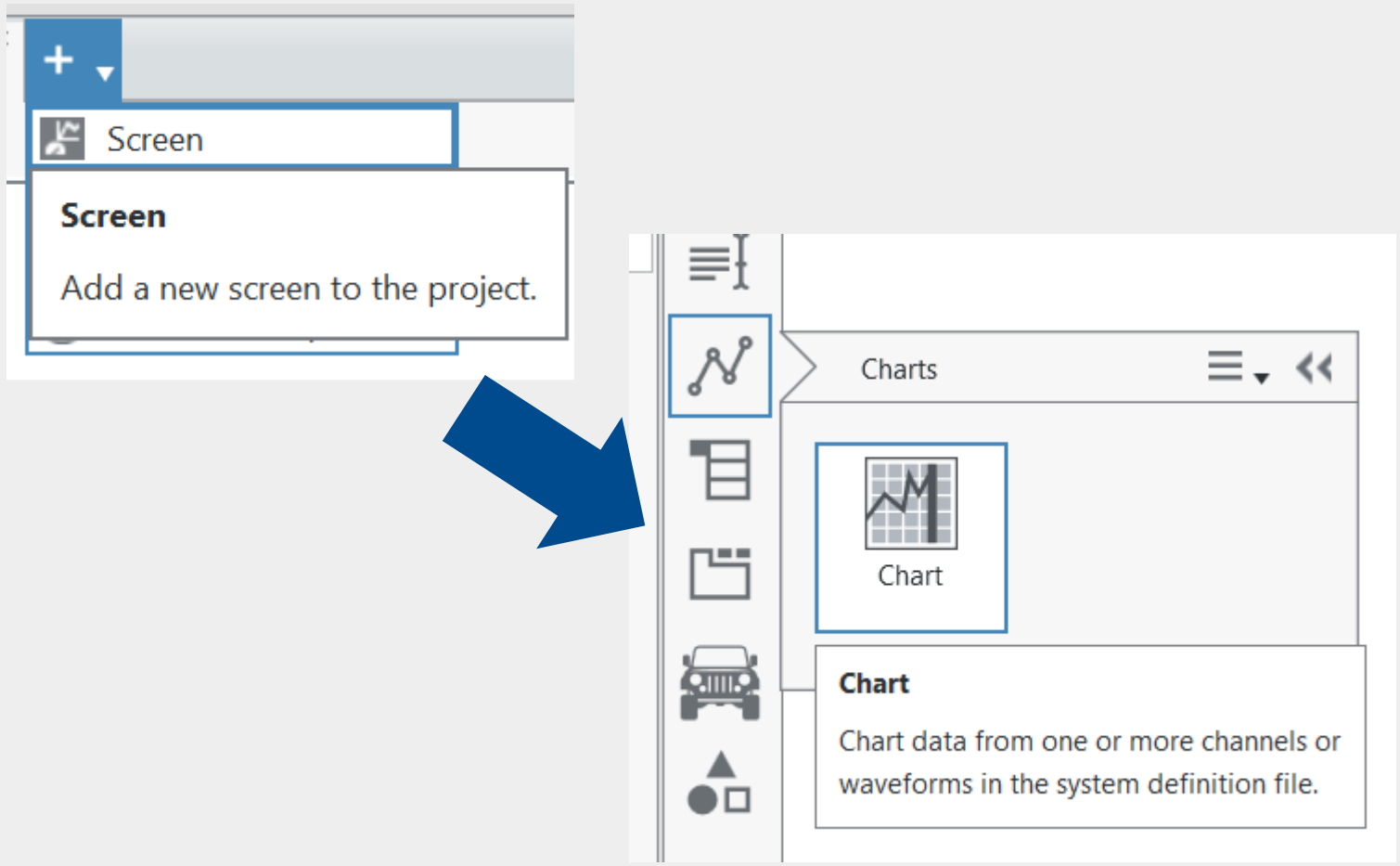
Screens

From the VeriStand Project

Add a Screen

Add a Chart

Create as many screens as needed

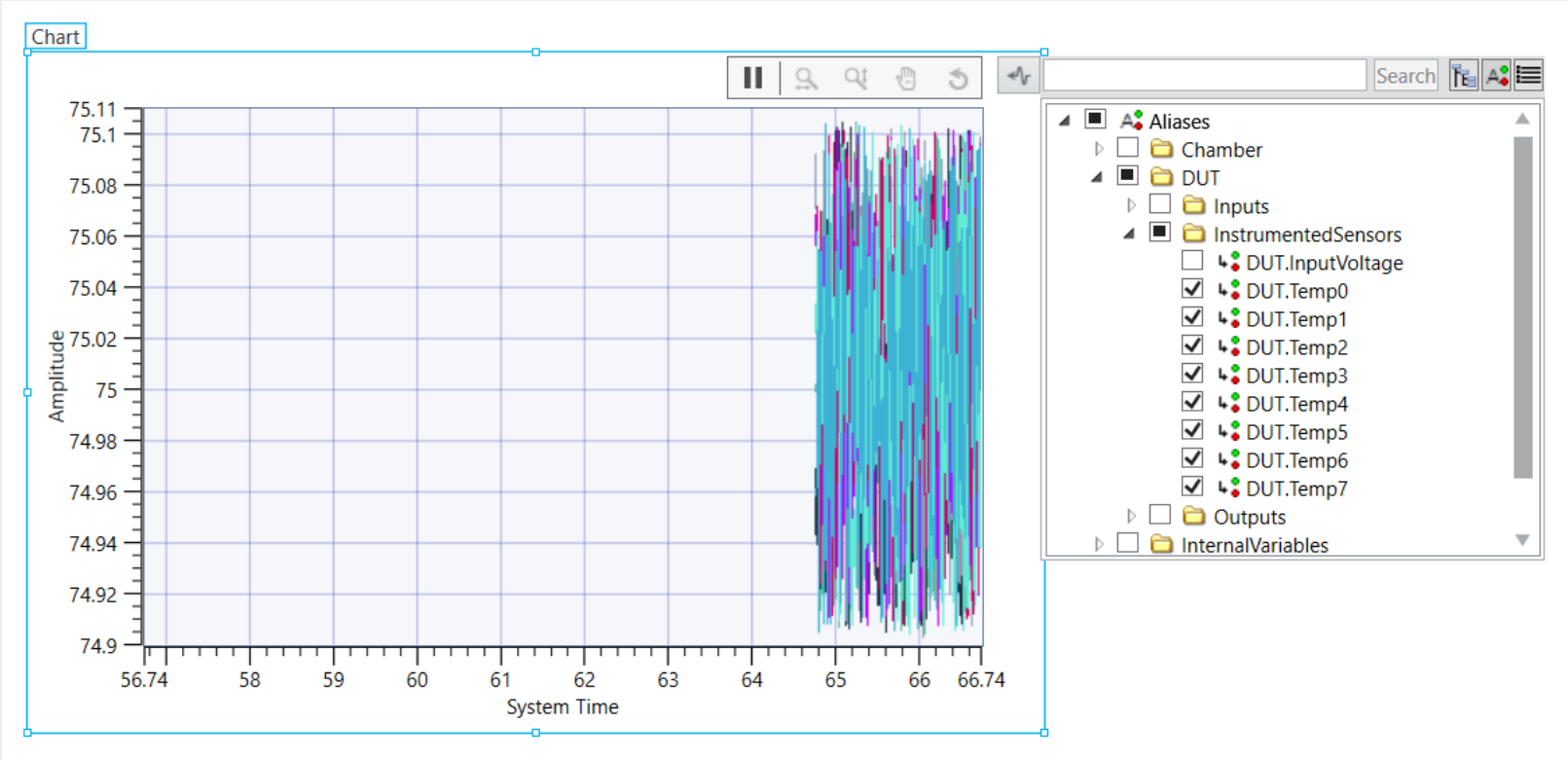


Configure GUI – Host Side UI

Screens

On the Chart

Select the channels to plot



Run the Demo!

Embedded Logger Project

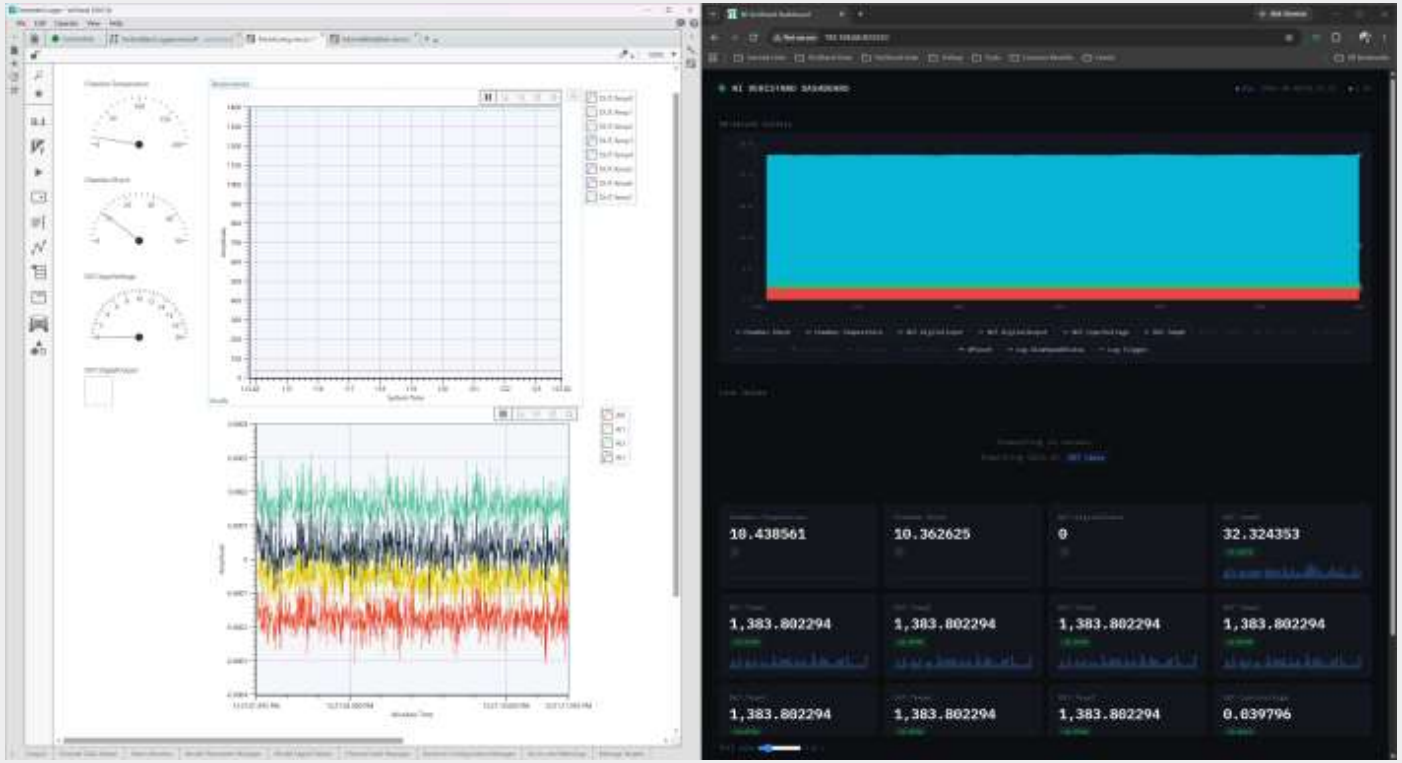
Open the Project

Review the configuration

Deploy the configuration

Show the Screens

On the InternalVariables screen, Set Log.Trigger to 1



Retrieve Data

Filezilla

Set the IP

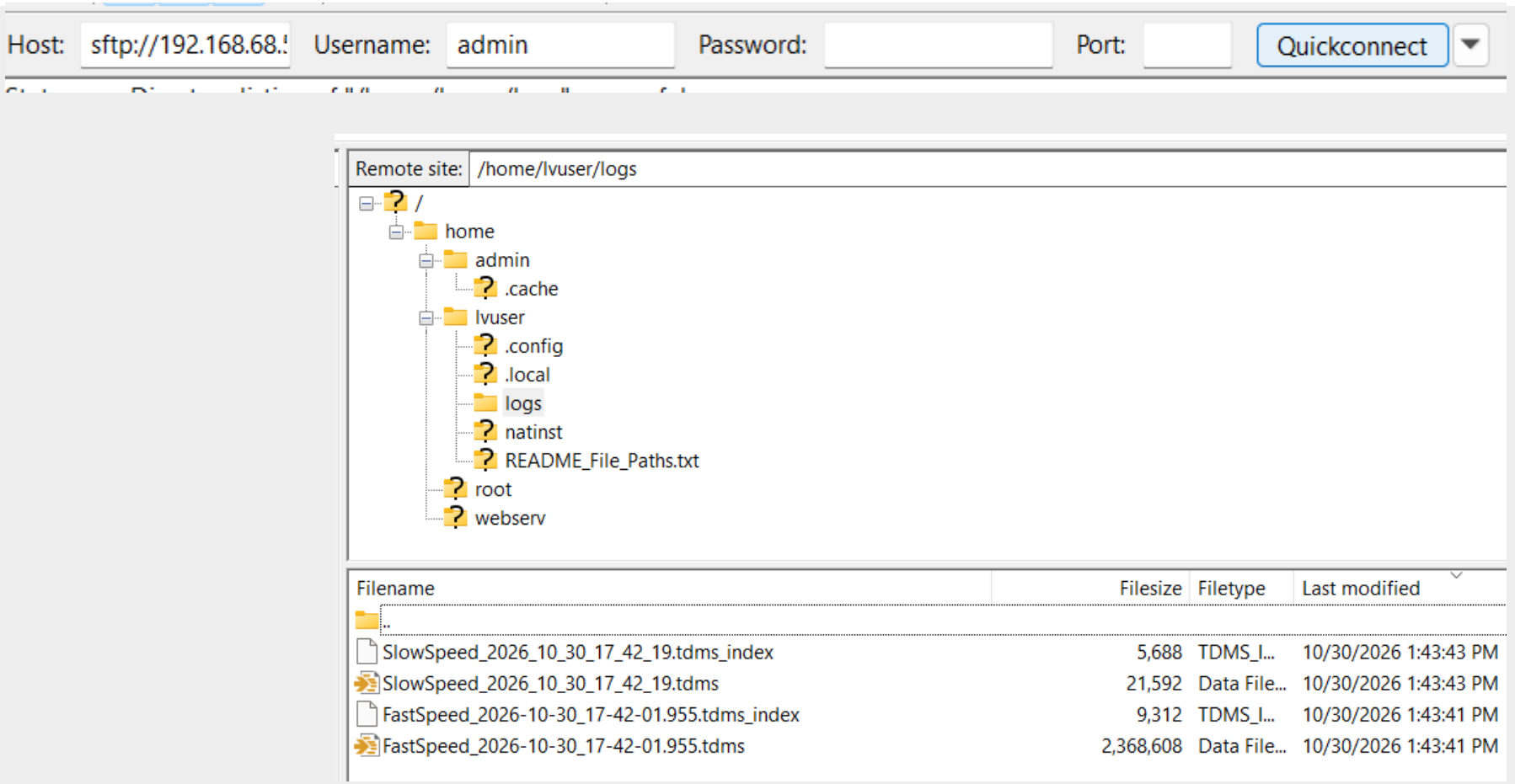
Set the Username

Set the Port

Connect

Navigate to /home/lvuser/log

Download Data



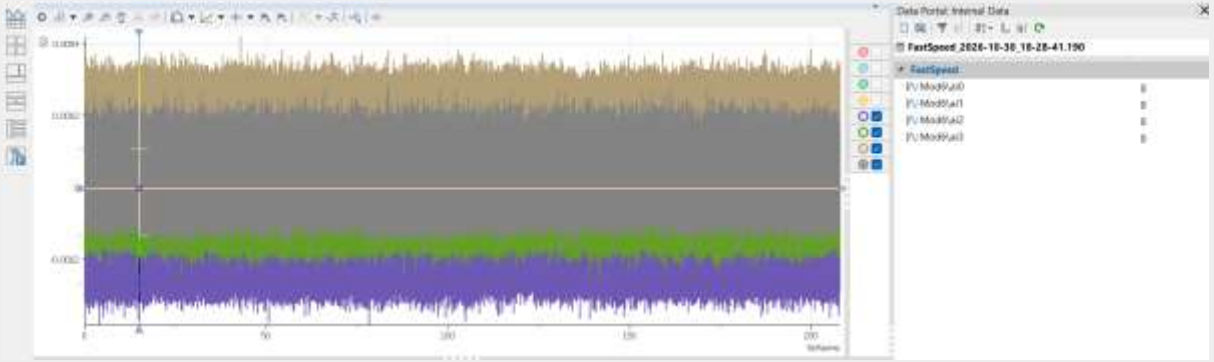
Display Logged Data

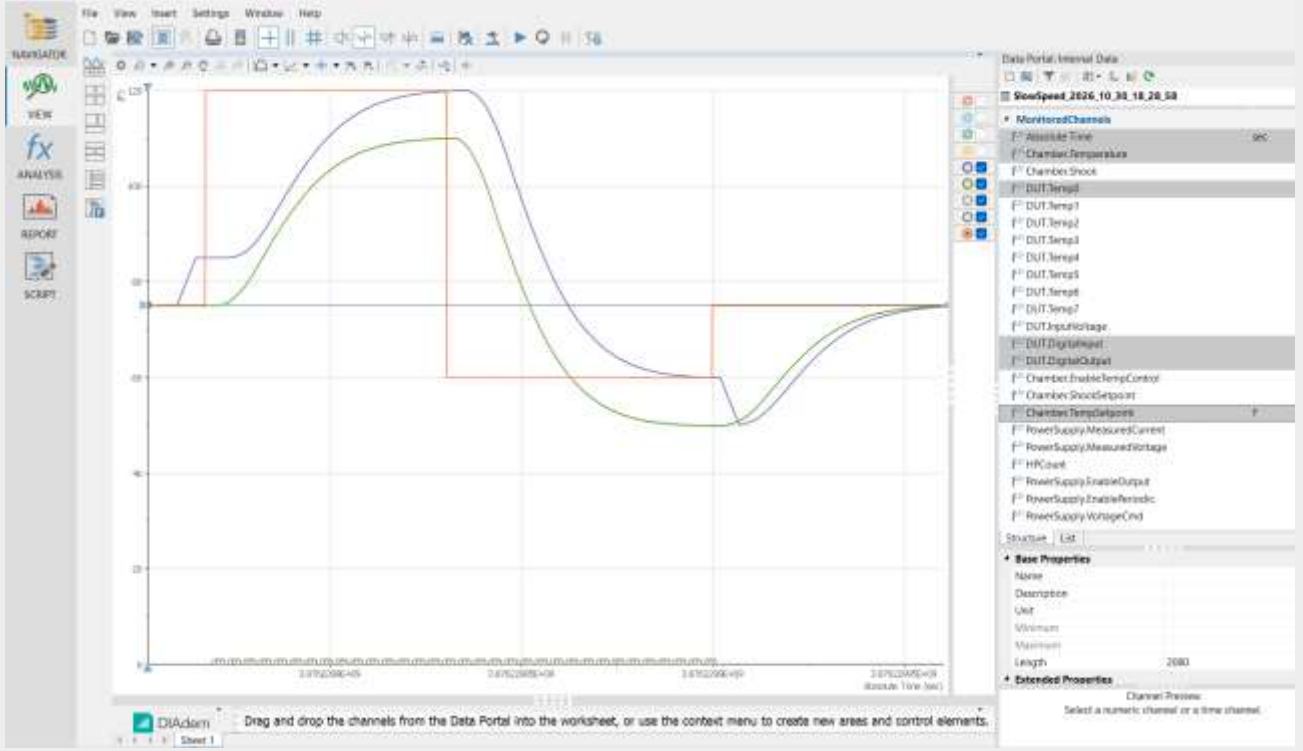
Diadem

Open the downloaded log in Diadem

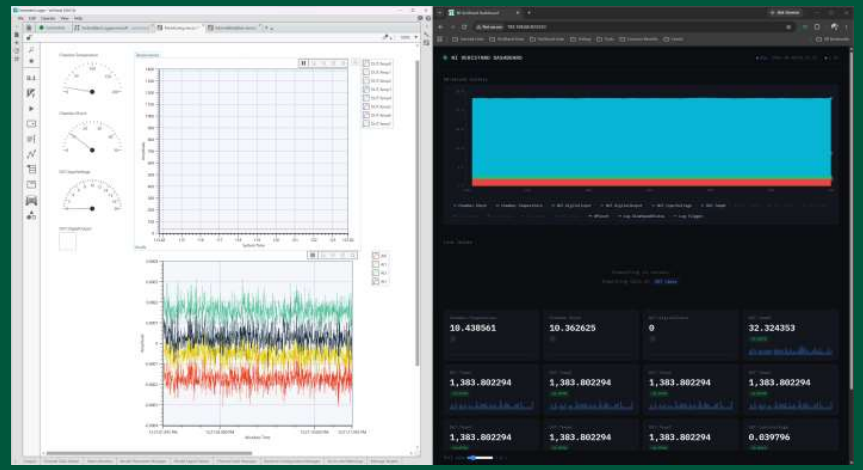
Plot a couple channels

Repeat with Slow & Fast speed data





Demo: NI VeriStand as Embedded Logger Summary

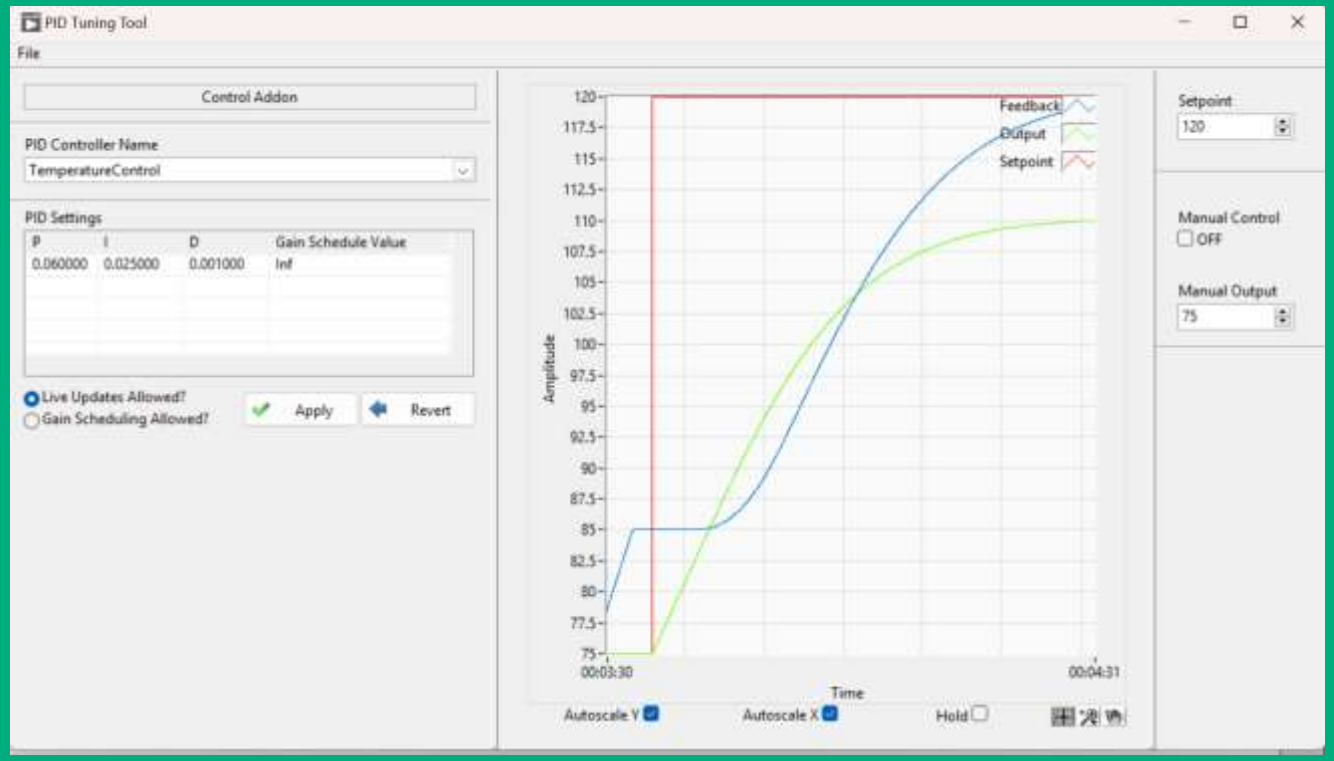


NO CODING REQUIRED

- Acquire High & Slow Speed Data
- Log all data
- Plot all the data (GUI)
- Publish all the data (webpage)
- Real Time Deployment & Startup

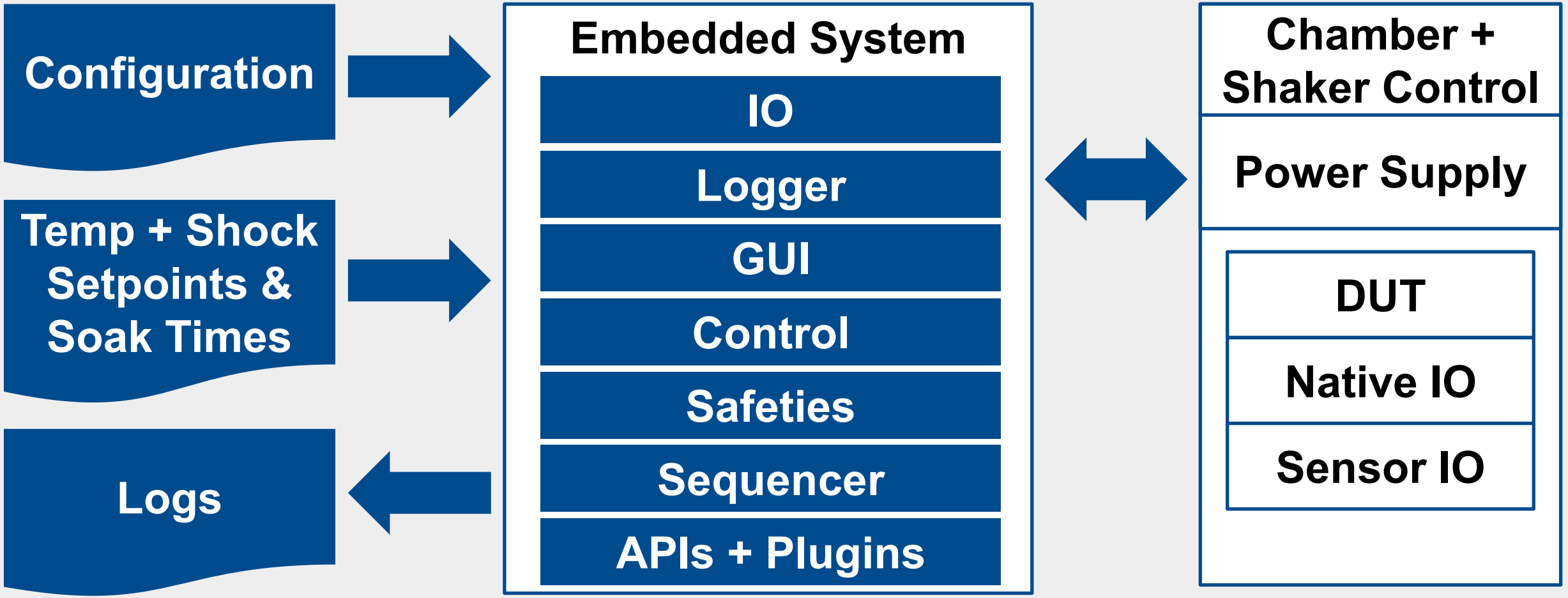
You Own a Configuration, Not Code!!

15 Minutes and Done!!



Demo: NI VeriStand as Embedded Control System

Demo Requirements



Demo: NI VeriStand as Embedded Control System

Input Channel List

DUT Instrumentation:

- 8 Thermocouples → NI 9213 (Mod1)
- 4 Accelerometers → NI 9234 (Mod3)

DUT IO

- Supply Voltage → NI 9221 (Mod2; ai0)
- 1 Digital Output → NI 9401 (Mod4; di0)

Chamber

- Temperature (C → Volt) → NI 9215 (Mod2; ai0)
- Shock (gs → Volt) → NI 9215 (Mod2; ai1)

Output Channel List

DUT Input

- 1 Digital Input → NI 9474 (Mod5; do0)

Chamber Control

- Enable Temperature Control → NI 9474 (Mod5; do1)
- Temperature Setpoint → NI 9263 (Mod6; ao0)
- Shock Setpoint → NI 9263 (Mod6; ao1)

3rd Party Power Supply

- Voltage → SCPI
- Current → SCPI
- Status → SCPI



Features - Simulation

- The Mathworks, Inc Simulink
- Esterel SCADE Suite
- C/C++
- Fortran/Ada
- MapleSim
- LabVIEW
- Rational Rhapsody
- and many others

Configure Simulation – Models

Build Simulation Model

Simulink:
https://www.youtube.com/watch?v=Ch8V3jR_vvo&list=PLYtAP08cxhvTGXErvD1U9tbdAwrXUTu9x&index=3

https://www.ni.com/docs/en-US/bundle/veristand/page/use-models-simulink.html

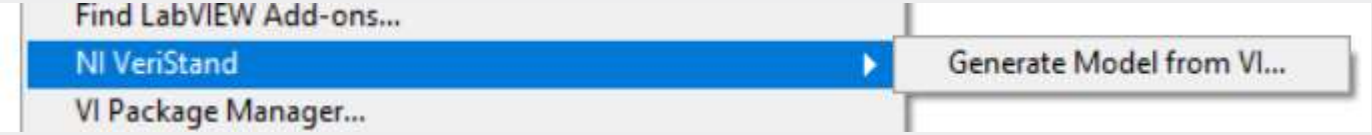
Configure Simulation – Models

Build Simulation Model

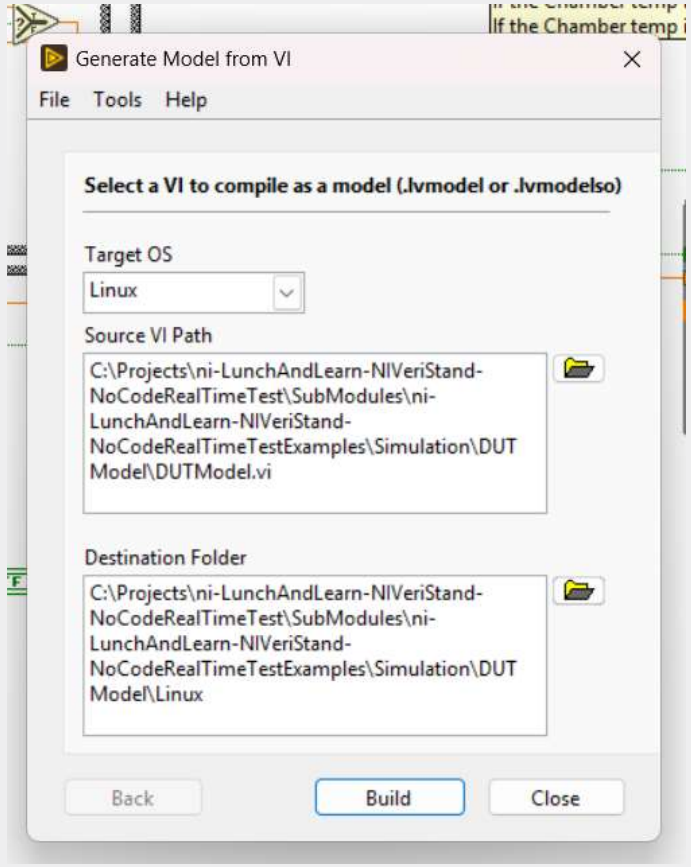
LabVIEW:
https://www.ni.com/docs/en-US/bundle/veristand/page/using-models-labview.html

From LabVIEW:

Tools → NI VeriStand → Generate Model From VI…



Examples at:
..\ni-LunchAndLearn-NIVeriStand-NoCodeRealTimeTestExamples\Simulation



Configure Simulation – Models

Add Models to Configuration

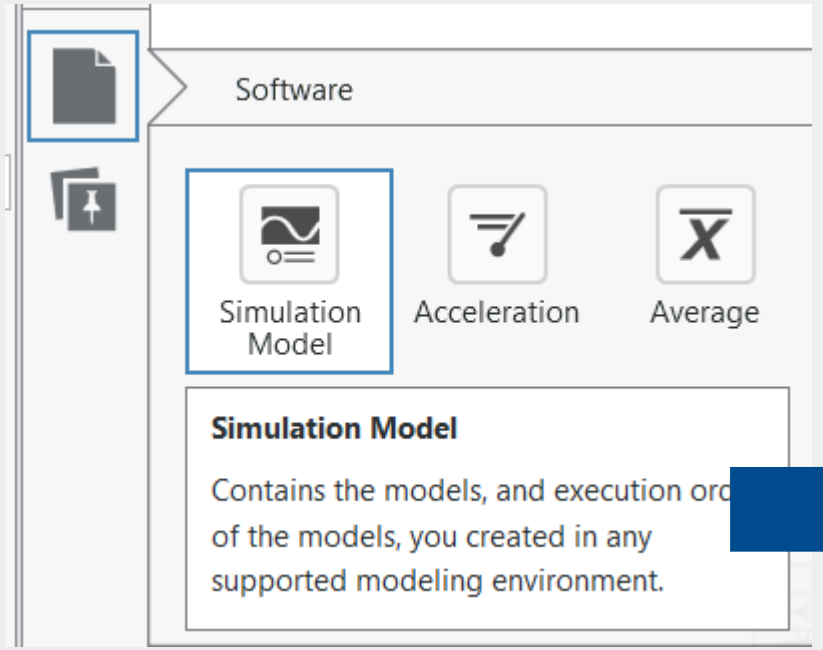
From Main Configuration Page

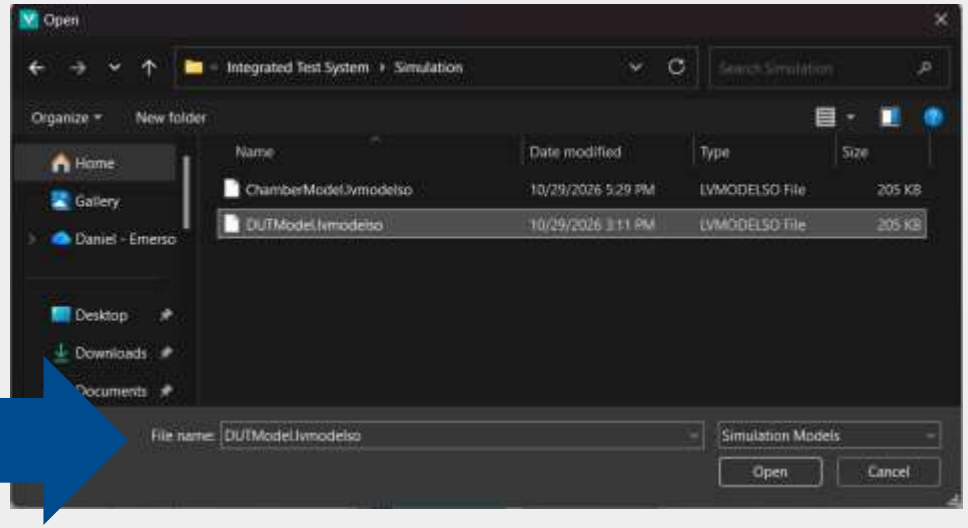
Expand Software

Select Simulation Model

Select compiled model on disk

Work the same for all model types





Configure Hardware – Aliases

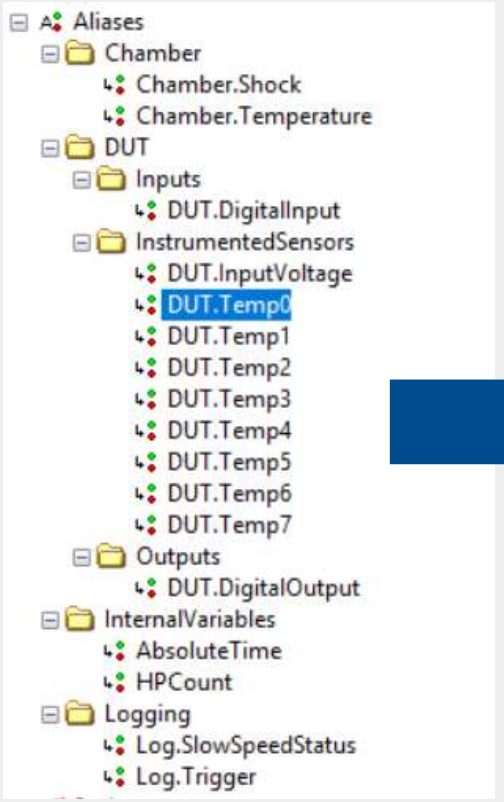
Alias Everything!!

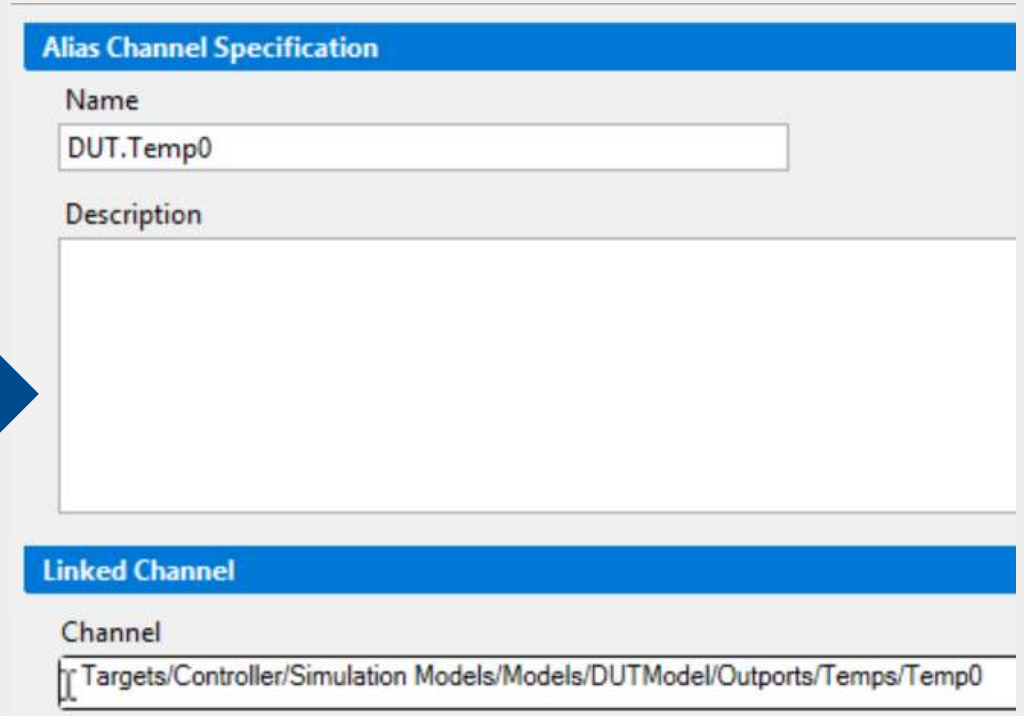
Swap Hardware IO for Model IO

Don’t change logging config

Don’t change GUI config

Don’t change Webpage config





Features – Manipulation

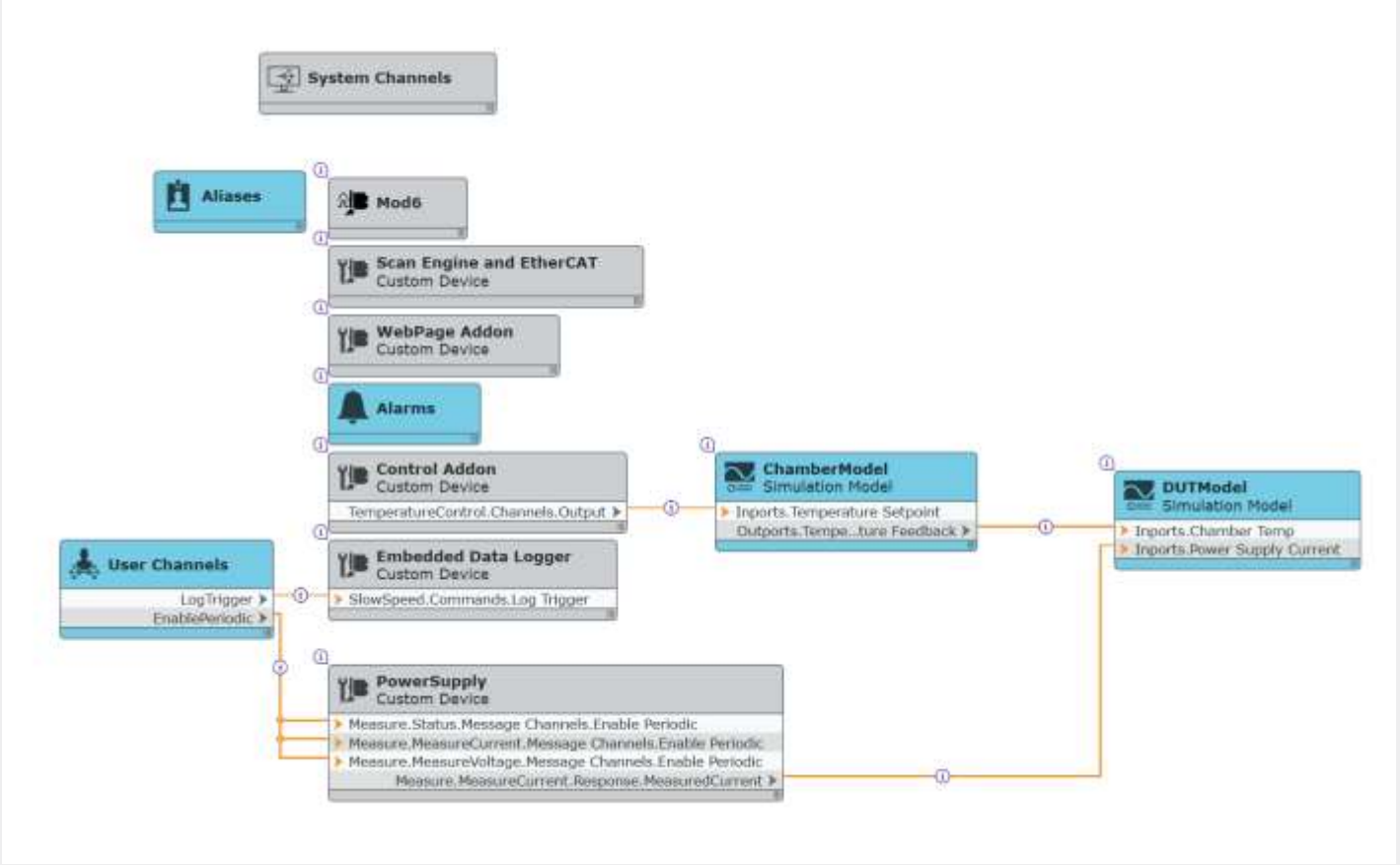
Mapping Channels

Channel Based Data Transfer Mechanism

Connect Channels to Anything (based on channel direction)

Examples:

- Connect IO to Models
- Connect single user Trigger to multiple trigger channels



Configure Control – PID

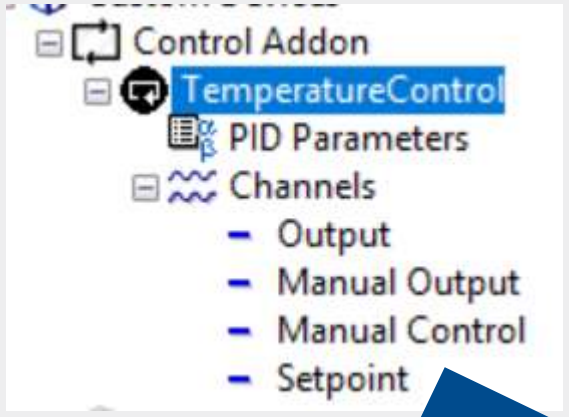
Control Addon

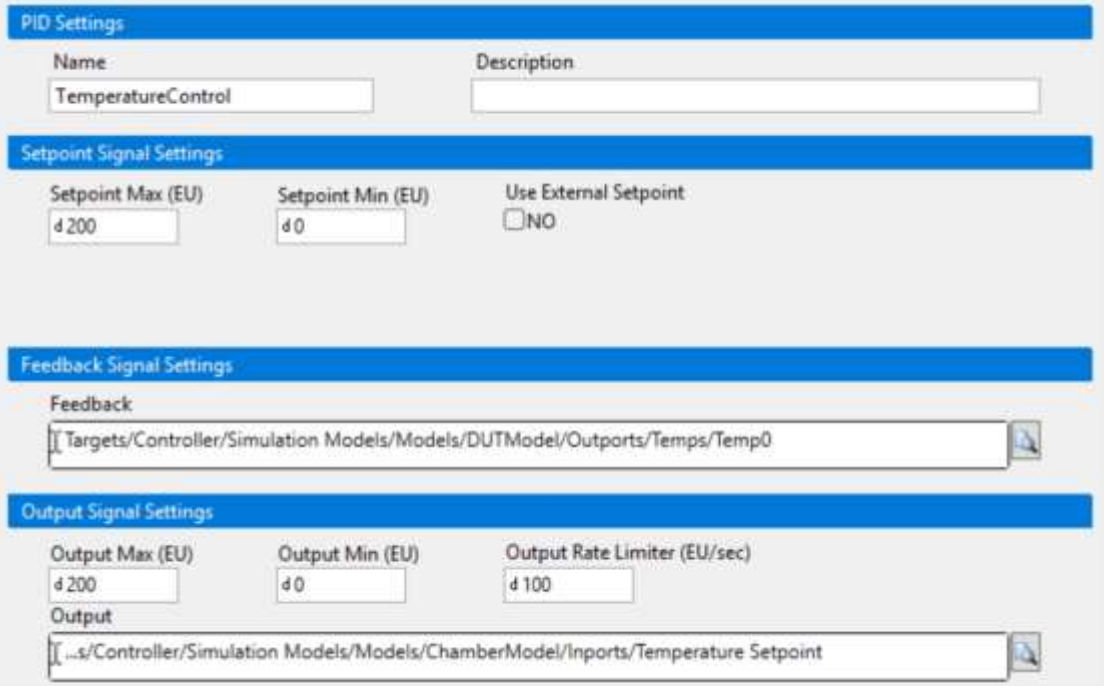
Add Control Addon

Add PID

On PID

- Select Feedback Channel
- Select Output Channel
- Set Max / Min Values
- Set Rate Limiter





Source Code:

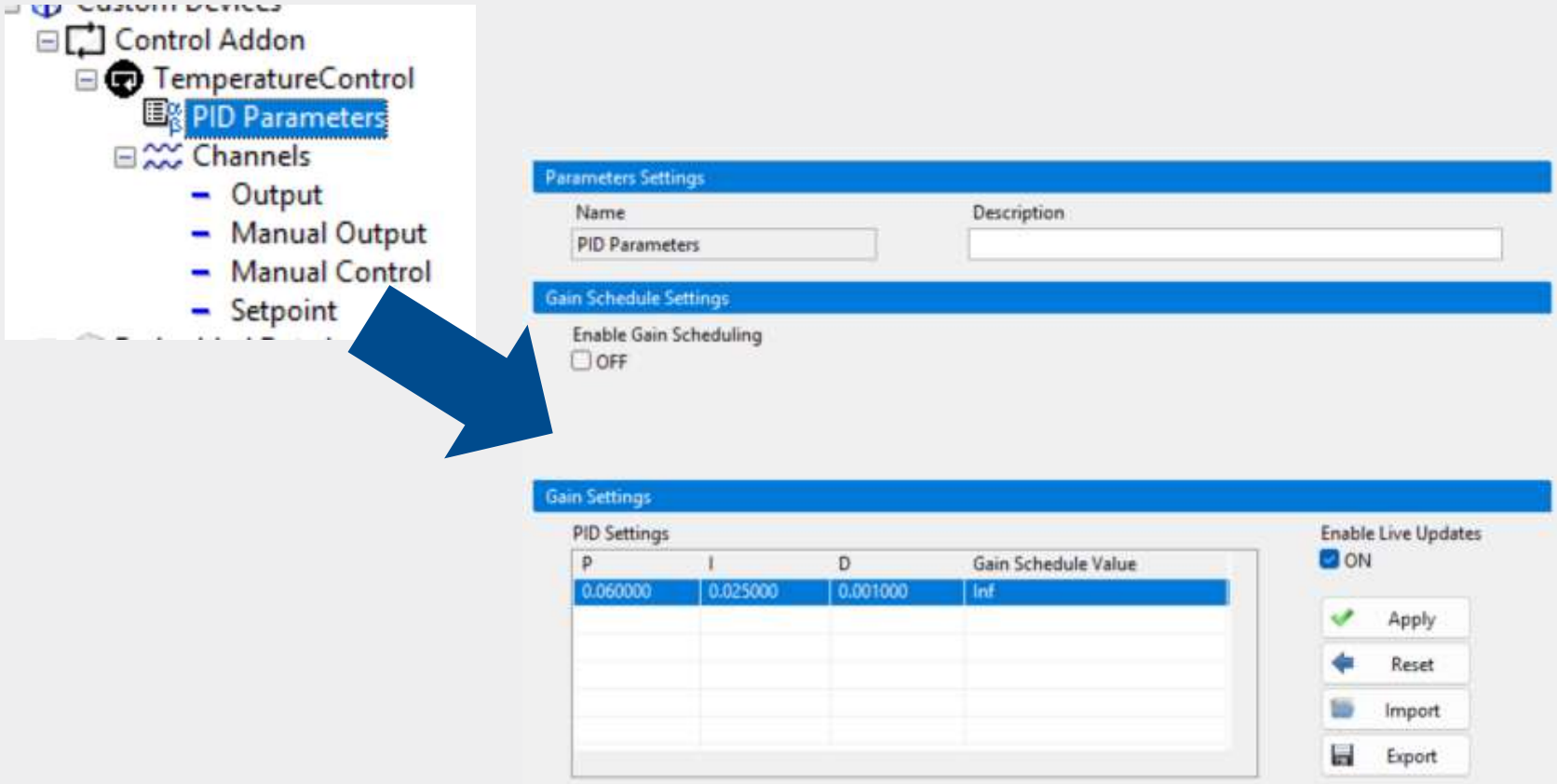
https://github.com/NIVeriStandAdd-Ons/Control-Custom-Device

Configure Control – PID

Control Addon

On PID Parameters

- Set PID Values
- Enable Live Updates to Tune

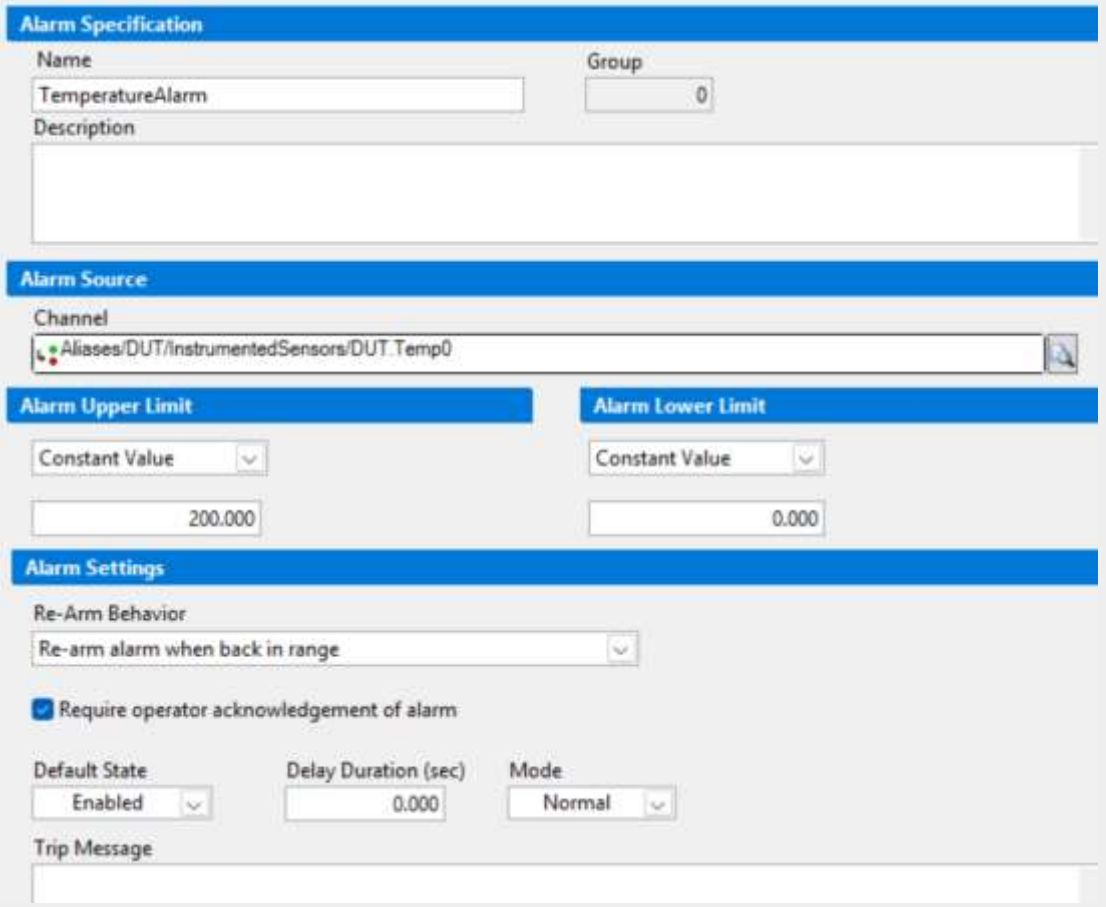


Features - Manipulation

Alarms

Check channel against limits

Trigger an action when limits are exceeded



Configure Safeties – Shutdown Procedure

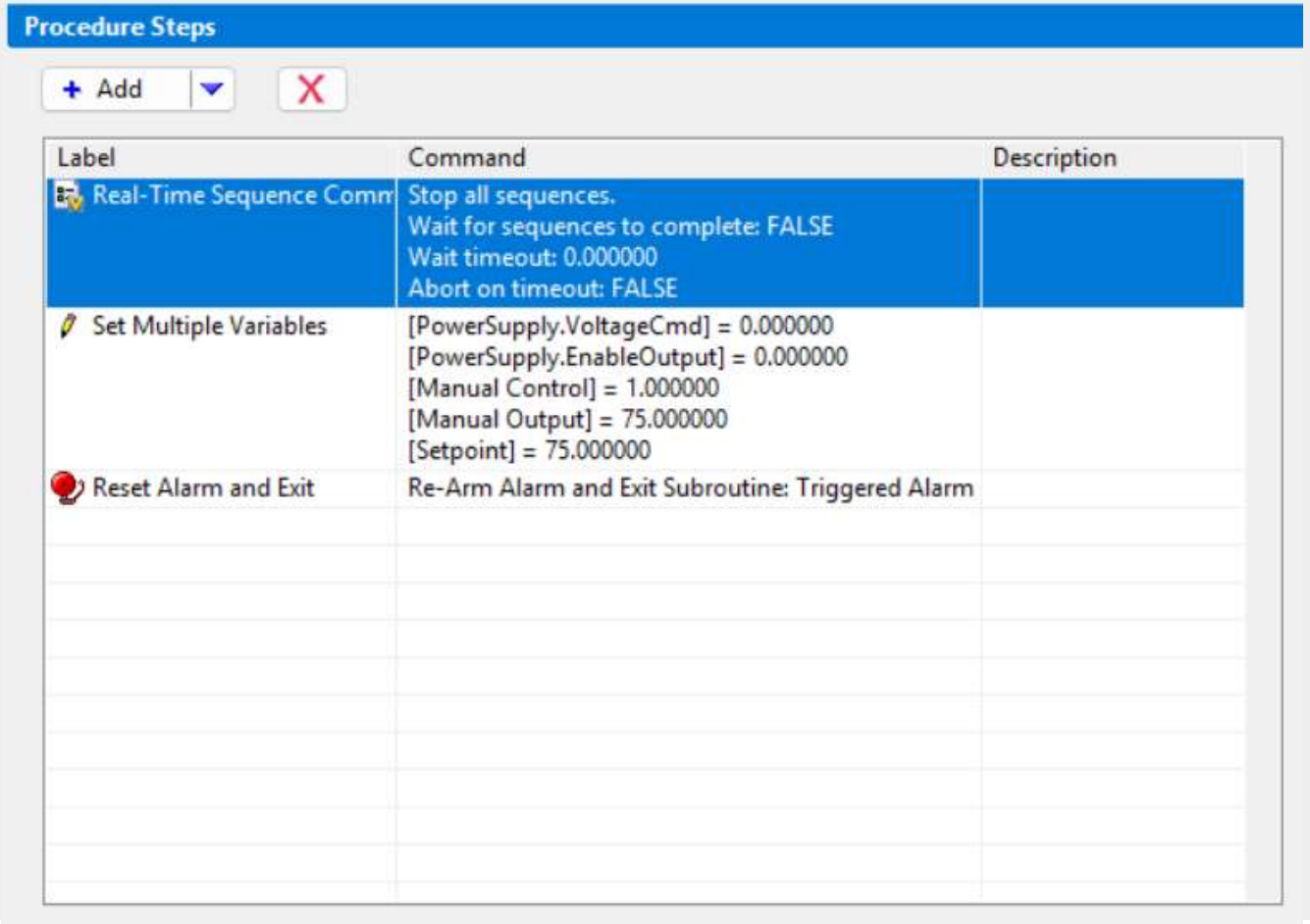
Create Procedure

Create a Procedure to safely shutdown system

Stop all running sequences

Set outputs to safe state

Reset Alarm and Exit



Configure Safeties – Alarming

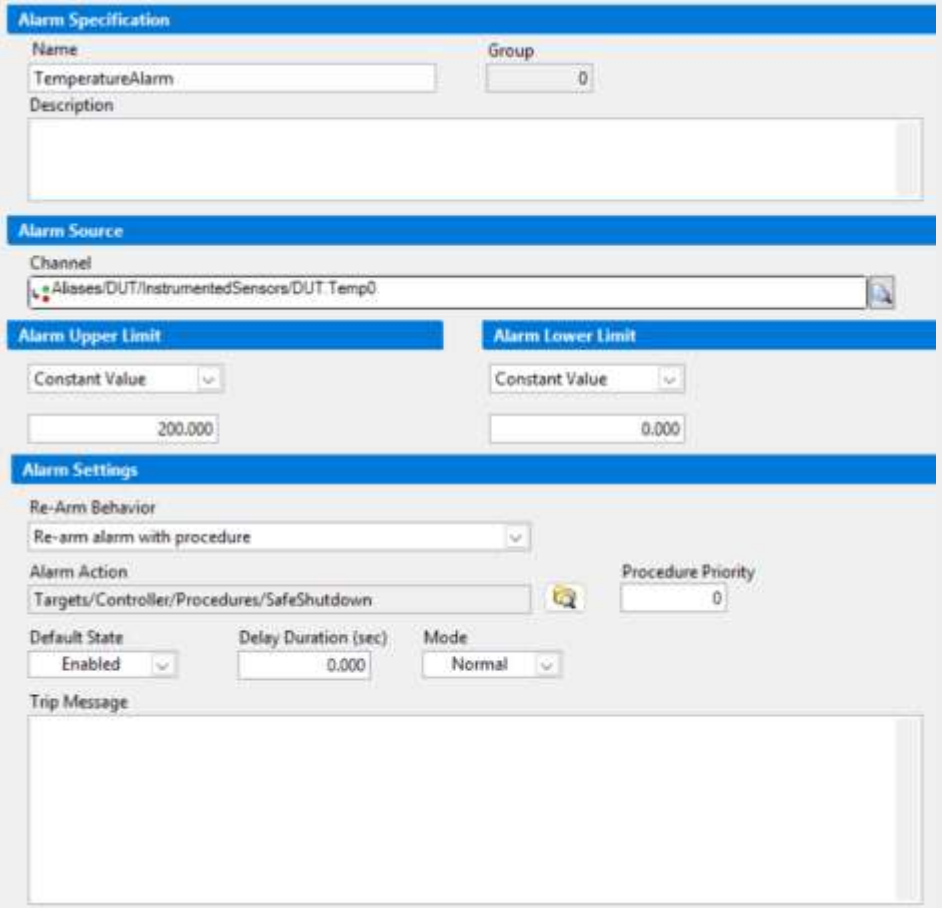
Add Alarm

Add an alarm

Select the channel source

Set limits

Select the Action (Procedure) to call when tripped



Configure Hardware – 3rd Party IO

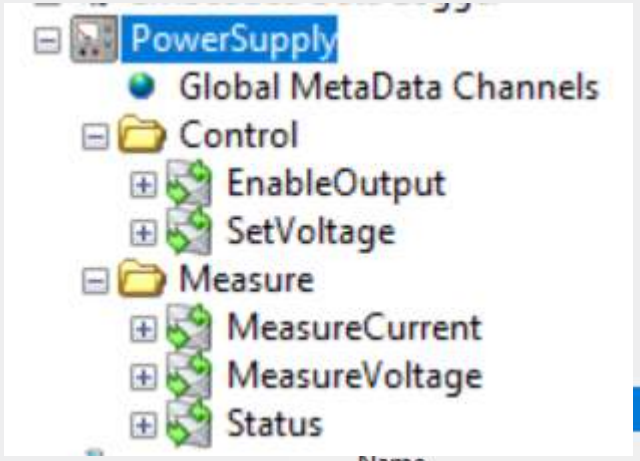
Instrument Addon

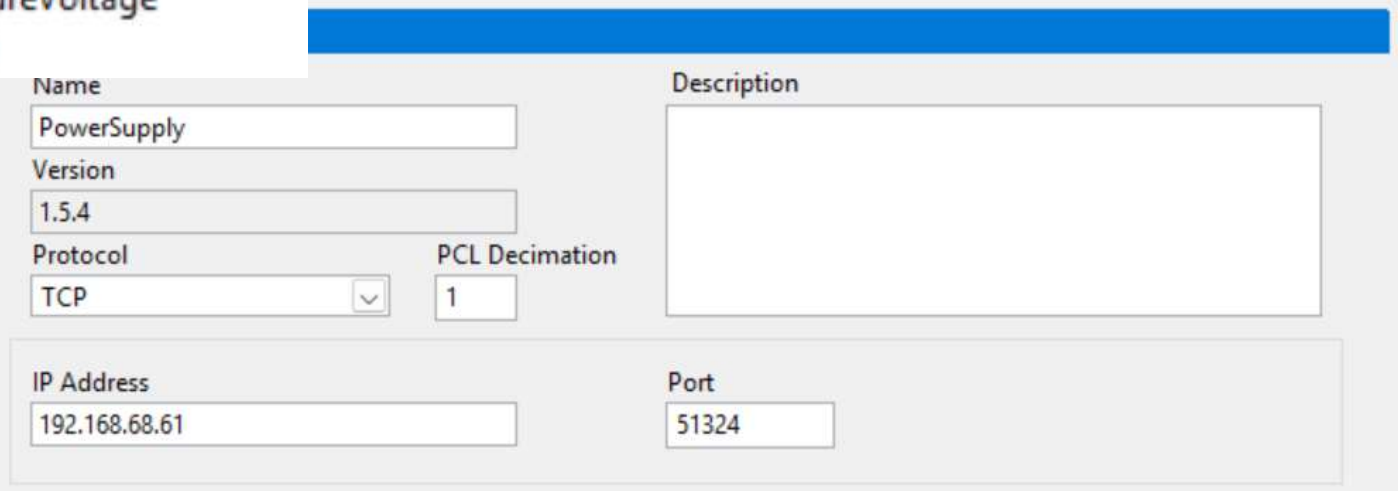
Configure SCPI Commands

Parses strings into/from data channels

Supports multiple hardware layers

- RS232
- TCP
- UDP
- GPIB





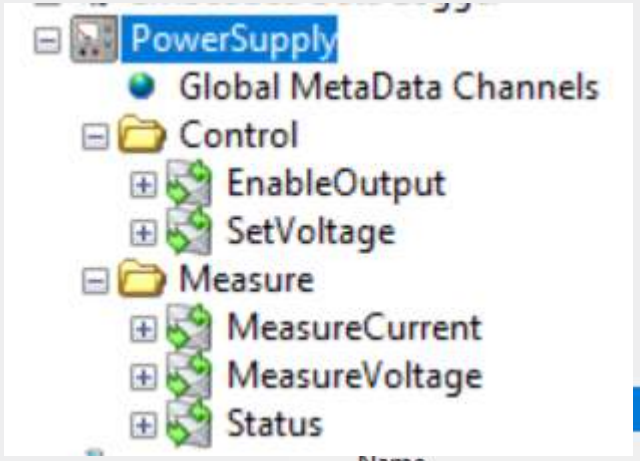
Configure Hardware – 3rd Party IO

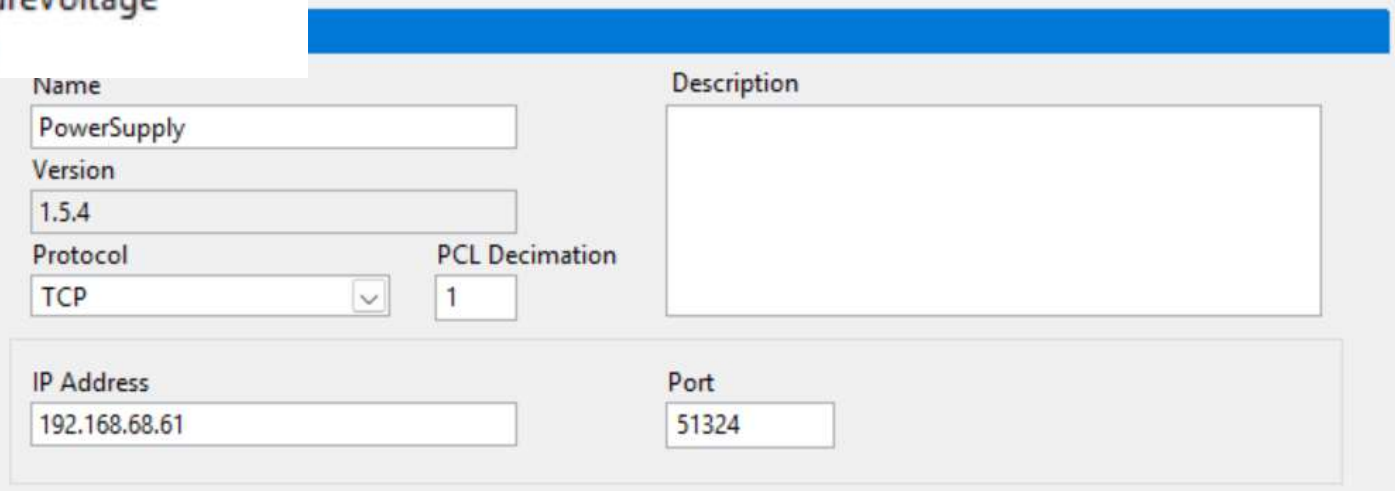
Instrument Addon

Add Addon

Pick your protocol

Set communication settings





Configure Hardware – 3rd Party IO

Instrument Addon

Add a Message Group

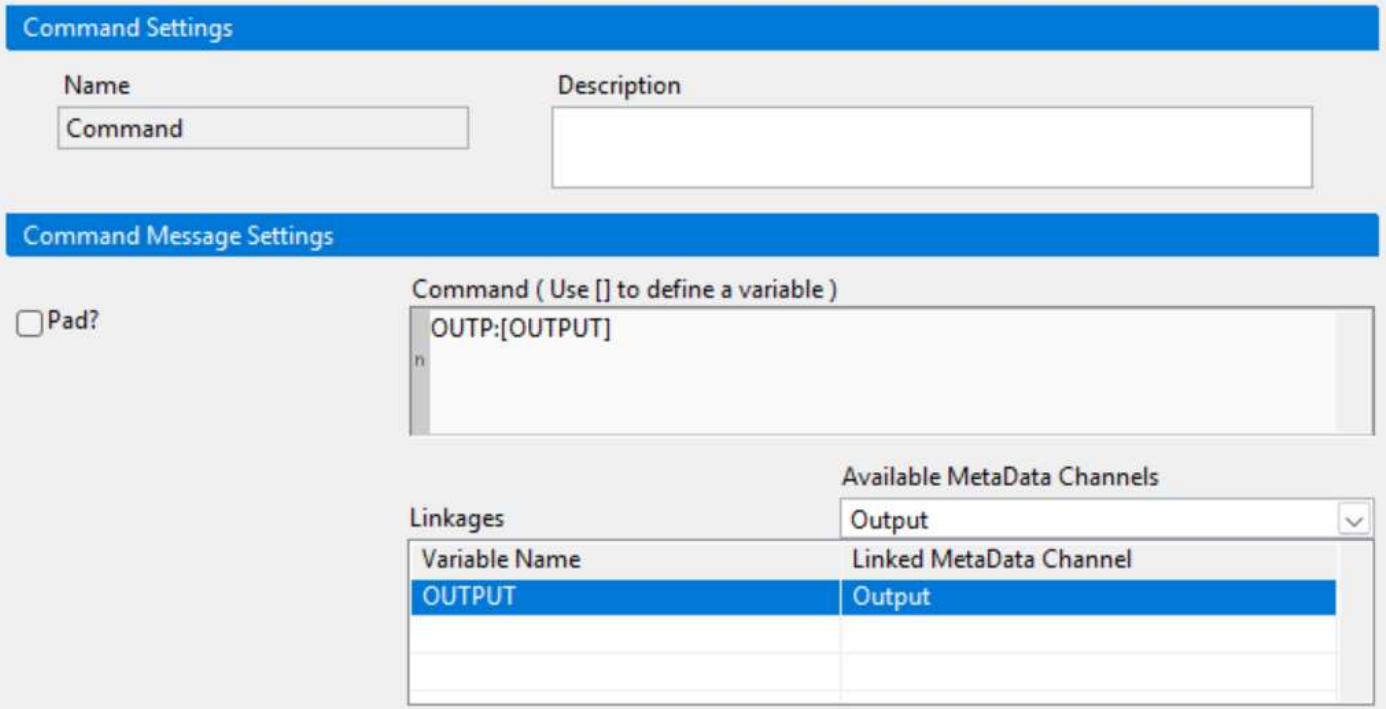
Add a Message

Configure the Cmd message

Parameterize the Cmd message

Add a Channel

Link the channel to the parameter

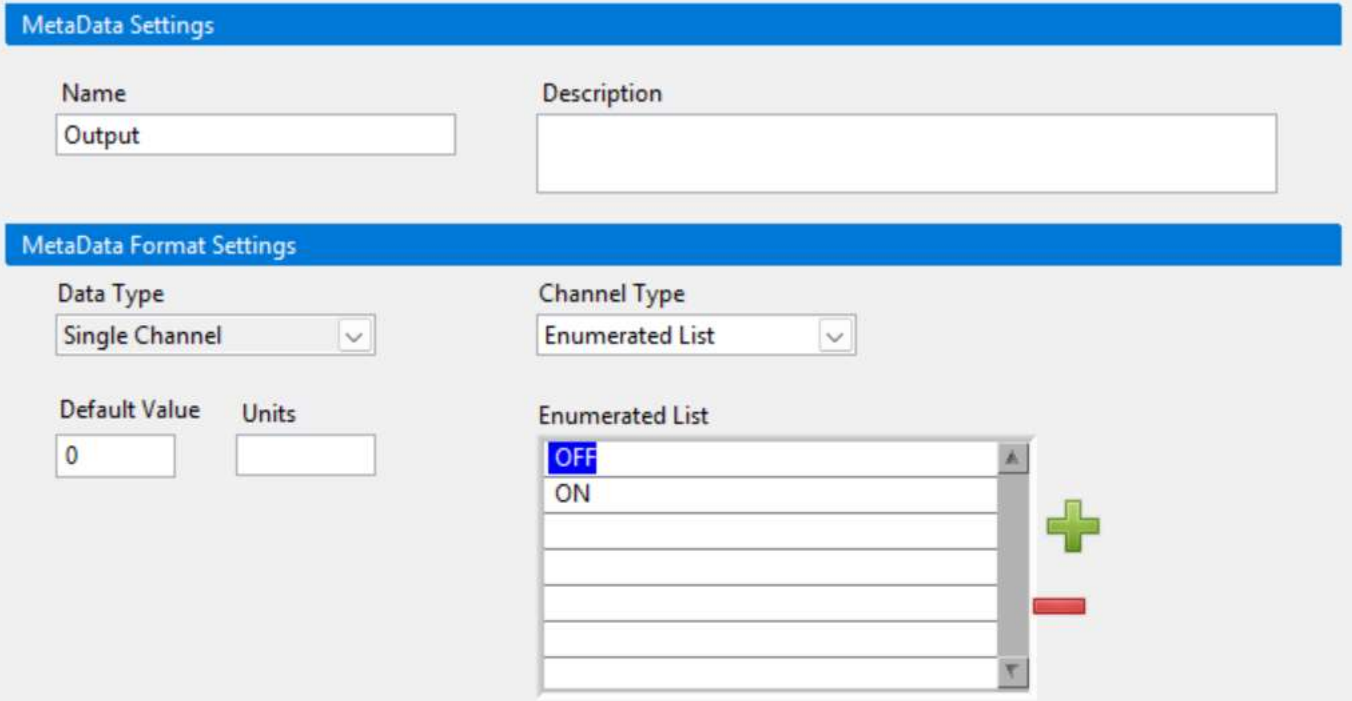


Configure Hardware – 3rd Party IO

Instrument Addon

On the added channel

Set the parsing Channel Type



Features – Automation

Procedures
(Low Level)

Simple procedures for common sequences

Designed in System Definition

Ex. Safe shutoff procedure

RT Sequence/Stimulus Profile
(Mid Level)

RT sequences that need to run in real time for control applications

- Designed in Sequence Editor
- Ex. Ramp up/down procedure

Sets of RT sequences run from host machine

- Designed in Profile Editor

TestStand / Python
(High Level)

Operator automation steps to control UI elements, through Veristand gateway (no direct RT control)

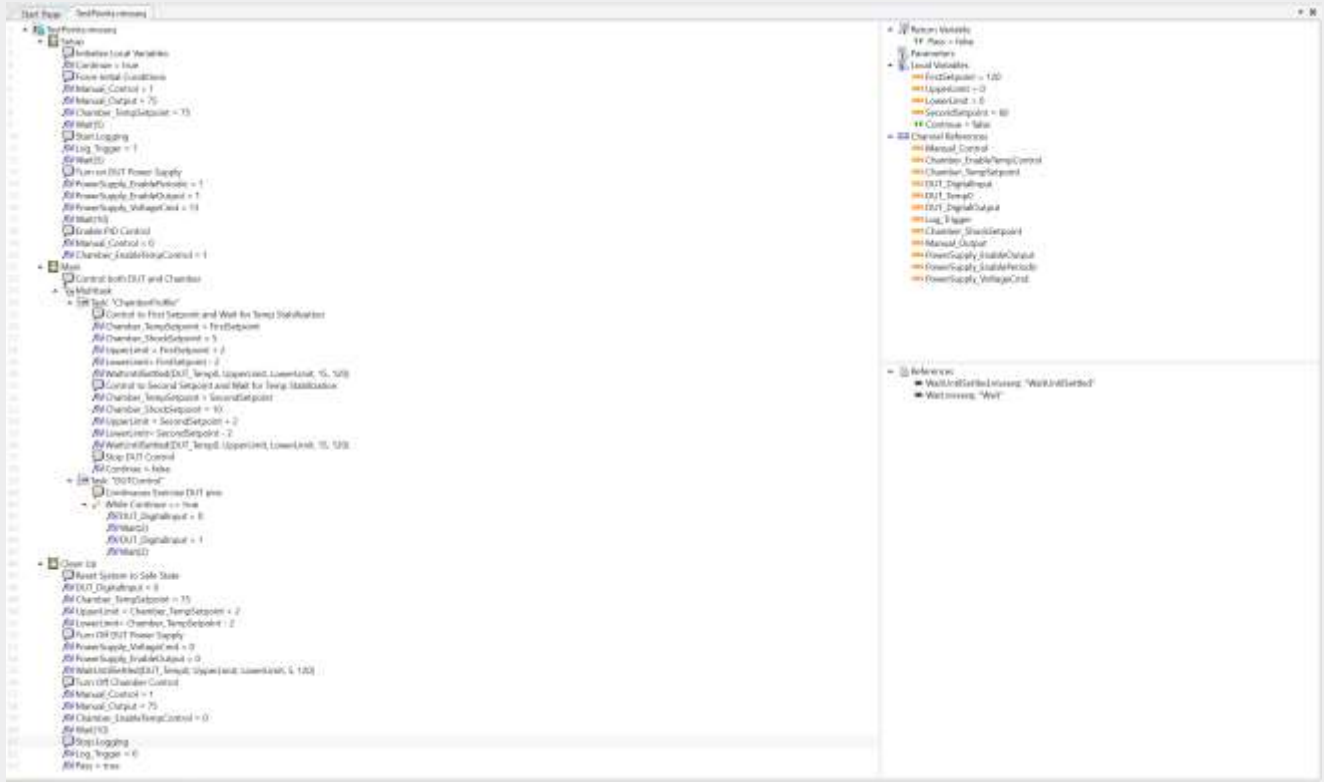
Configure Automation – RT Sequences

Stimulus Profile Editor

Create a new Sequence

Import Channel References (Alises)

Write and validate your sequence



Configure Automation – RT Sequences

Stimulus Profile Editor

From a Screen

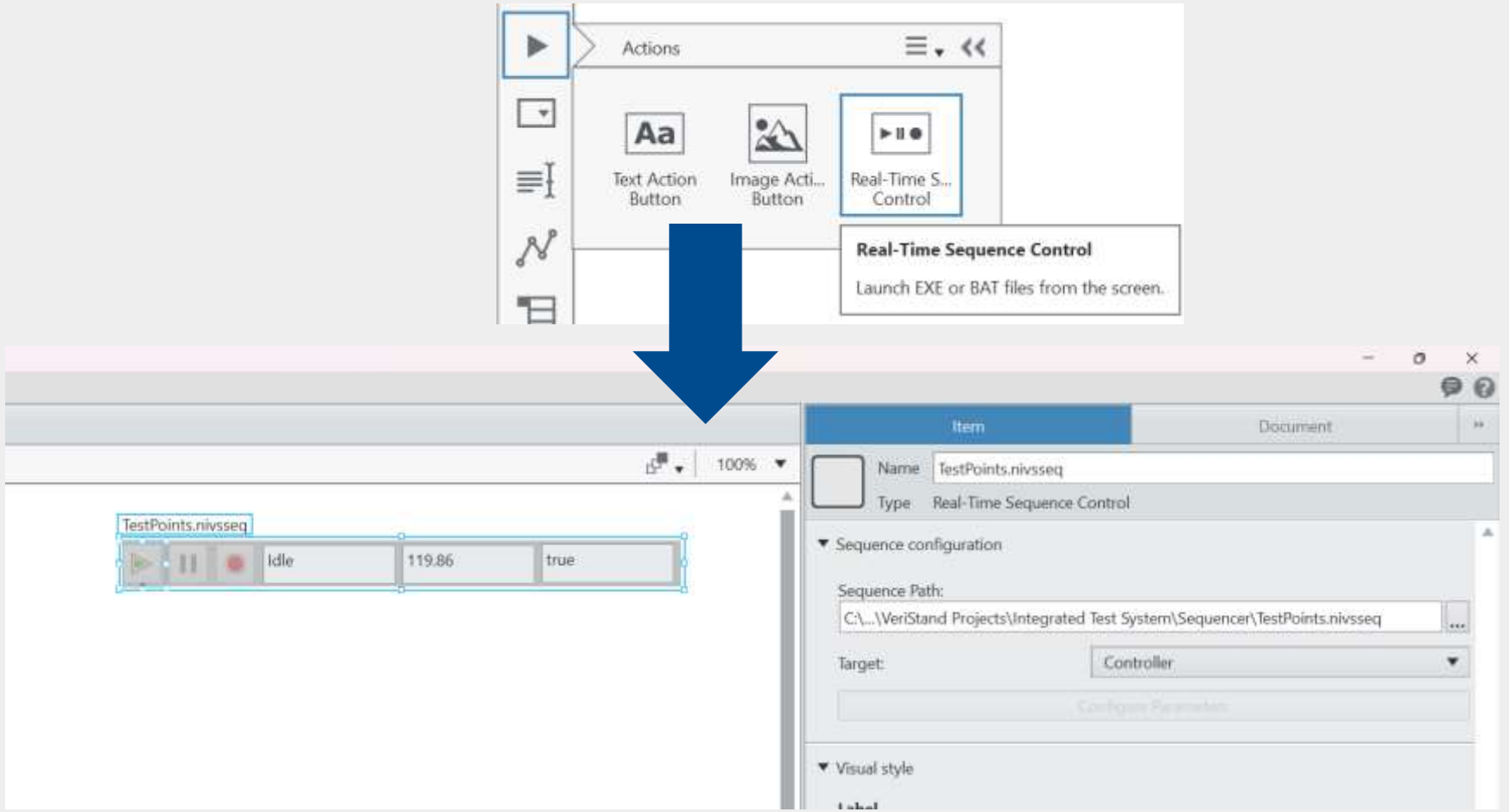
Select Actions

Select Real-Time Sequence Control

Add RT Sequence UI Object

Select RT Sequence Path

Run



Run the Demo!

Integrated Test System Project

Open the Project

Review the configuration

Check Instrument Addon IP matches your Host

From Project Files, Open Simulation → Power Supply.vi

Check Power Supply IP Matches your Host

Run Power Supply.vi

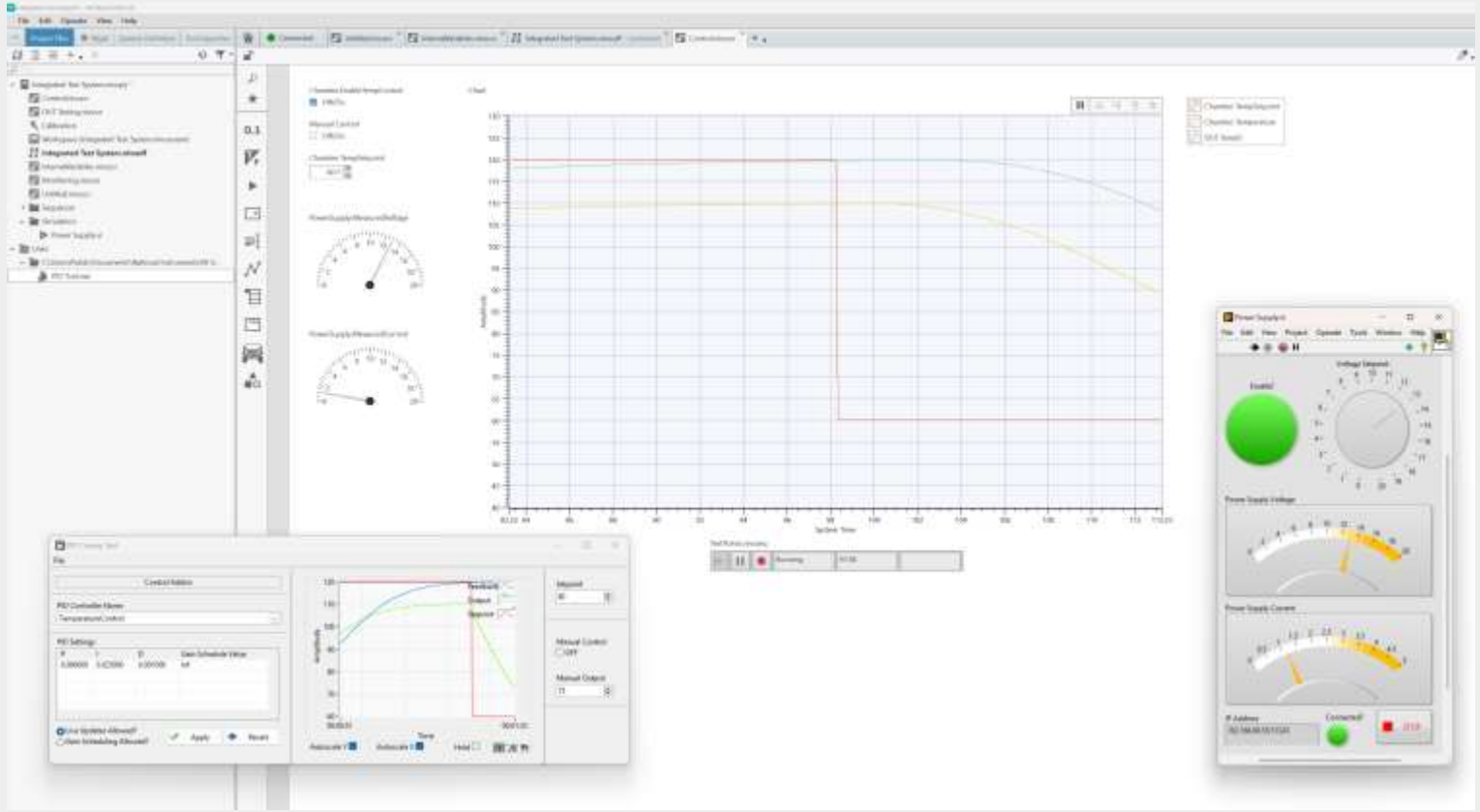
Deploy the configuration

Open PID Tool.exe from Links

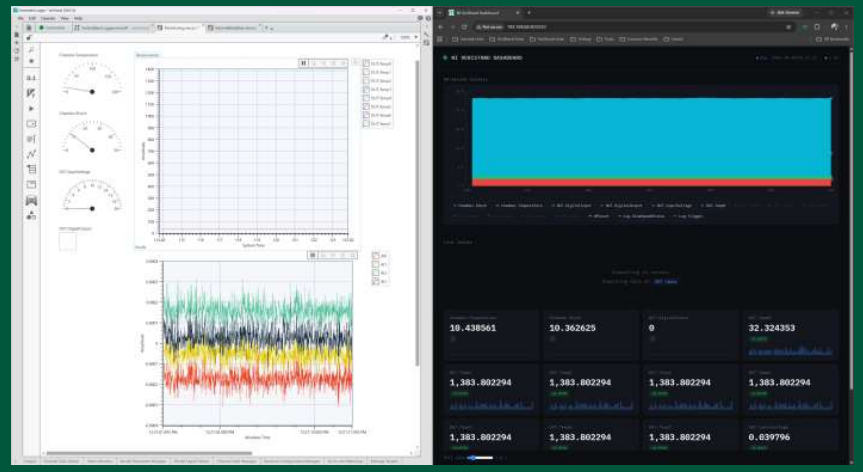
Connect Tool to addon (File→Configure→Control Addon)

Show the Screens

On the Control screen, Run the sequence



Demo: NI VeriStand as Embedded Control System Summary

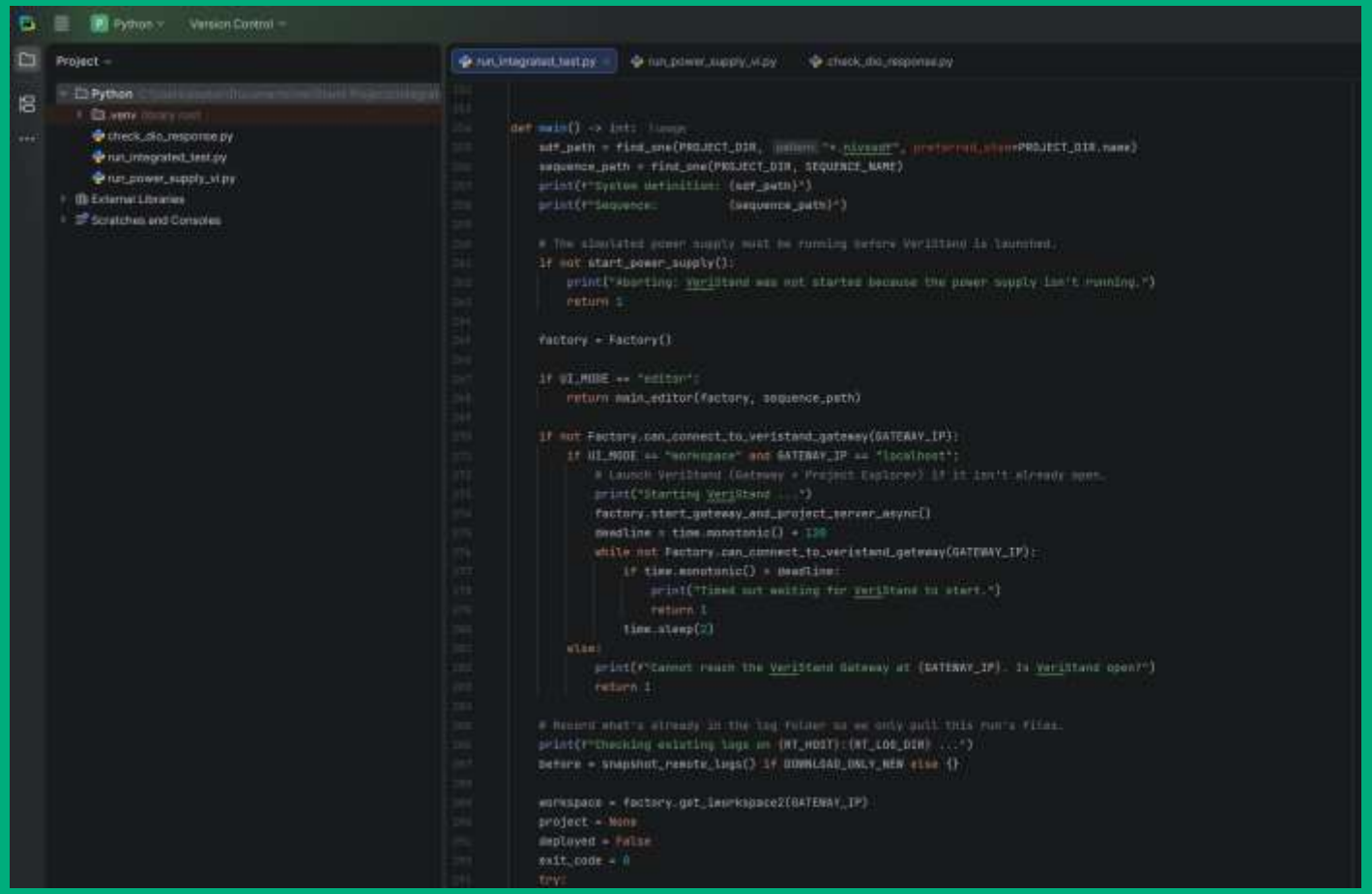


NO CODING REQUIRED

- Did all the previous things
- Simulated a system
- Controlled a system
- Set safeties and shutdown procedures
- Automated a test

You Own a Configuration, Not Code!!

15 Minutes and Done!!



Demo: NI VeriStand as Integrated Test System

VIBE CODE TIME!!

EMERSON | NI

Features – Extensions (Custom Devices)

Veristand Custom Devices



CUSTOM FUNCTIONALITY



BUILT IN LABVIEW



OPEN SOURCE DEVICES AVAILABLE

Features – Extensions

Options

Models	Executes custom code directly in the primary control loop. Written in LabVIEW or Simulink. Must finish execution within the loop rate. Great for control logic, calculations, state based logic, simulation algorithms, etc. Much simpler than custom devices but also more limited.
Custom Devices	Full featured, first-class extension of NI VeriStand configuration and execution engine. Written in LabVIEW (can wrap Python / C). Executes in primary control loop or asynchronously. Asynchronous allows for non-deterministic operations and is not required to finish in loop rate. Best for complicated algorithms, third-party IO, slow speed operations (logging, SCPI, etc), etc.
Workspace Tools	Executes custom GUIs within the host Workspace / GUI. Written in LabVIEW. Template provides access to deployed VeriStand configuration.
Workspace Services	Executes custom services within the host Workspace. Written in LabVIEW. Template provides access to deployed VeriStand configuration.

Examples from this Project

Custom Devices

- Control Addon (https://github.com/NIVeriStandAdd-Ons/Control-Custom-Device)
- Web Page Addon (https://github.com/NIVeriStandAdd-Ons/NI-Veristand-Webpage-Addon)
- Instrument Addon (R&D Built in)

Models

- DUT (..\Simulation\DUT Model)
- Chamber (..\Simulation\Chamber Model)

Tools

- PID Tuning Tool (https://github.com/NIVeriStandAdd-Ons/Control-Custom-Device)

Features - Interoperability

Options

Execution API	Automates the operation of an VeriStand application on the target. For example, you can **read and write channel data**, control running models, configure alarm states and read data from alarms, and access Workspace tools.
System Definition API	Automates the operation and configuration of a system definition file. This API performs the same operations as configuring the file in the System Explorer window.
Stimulus Profile Definition API	Automates the operation and configuration of stimulus profiles. This API performs the same operations as configuring stimulus profiles in the Stimulus Profile Editor.
Real-Time Sequence Definition API	Automates the operation and configuration of real-time sequences. This API performs the same operations as configuring real-time sequences in the Stimulus Profile Editor.
Data Types API	Represents data types and resources used by VeriStand.

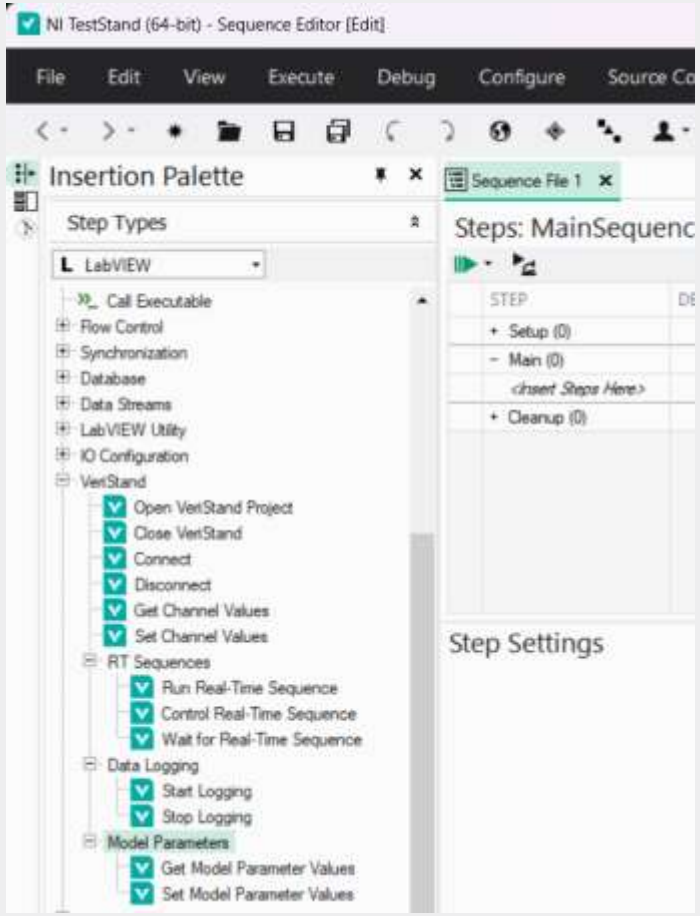
NI VeriStand TestStand API

Open / Close Project

Set / Get Channels (Non-deterministic sequencing)

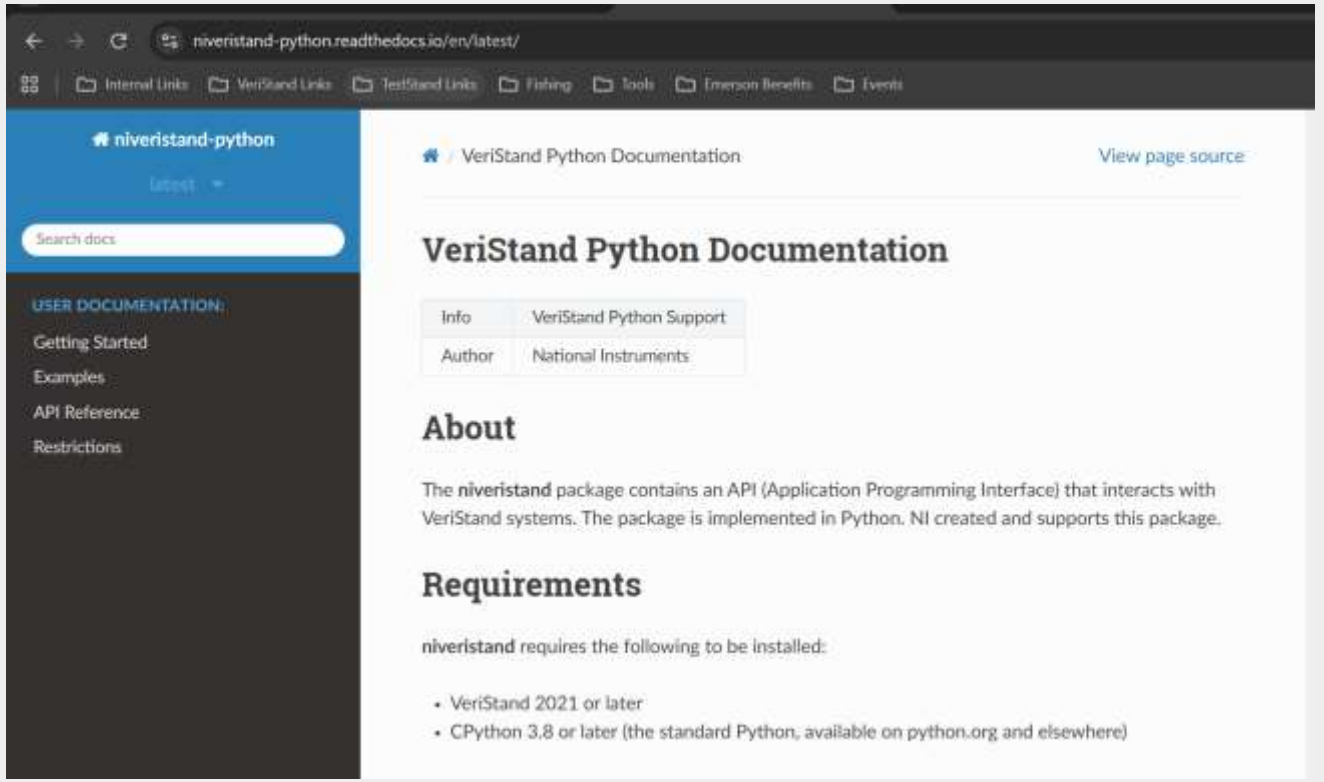
Run Sequences

Host Side Logging



NI VeriStand Python API

https://niveristand-python.readthedocs.io/en/latest/



NI VeriStand Python API

Prompt

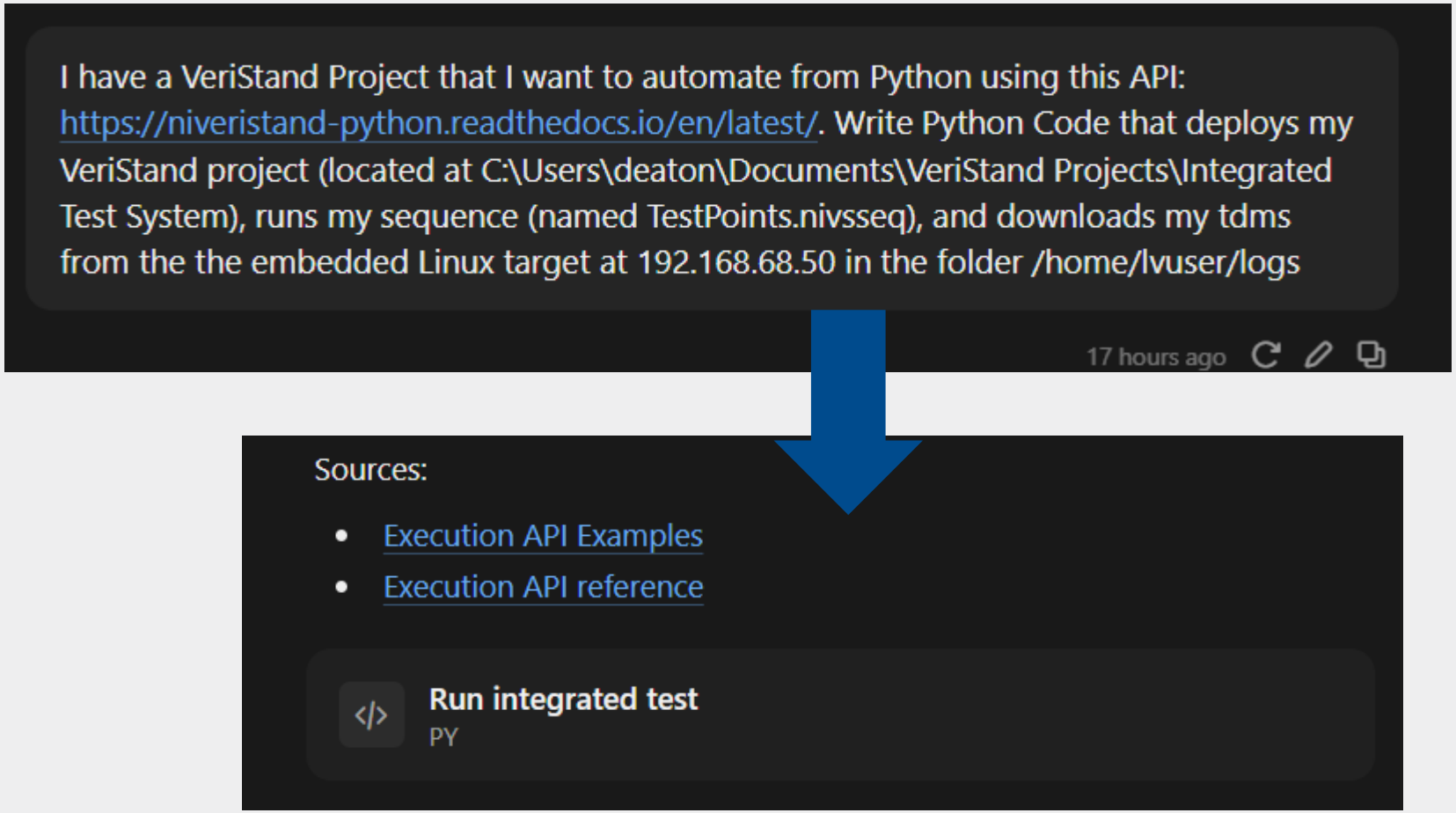
Run my Project

Run my sequence

Download resulting TDMS Data

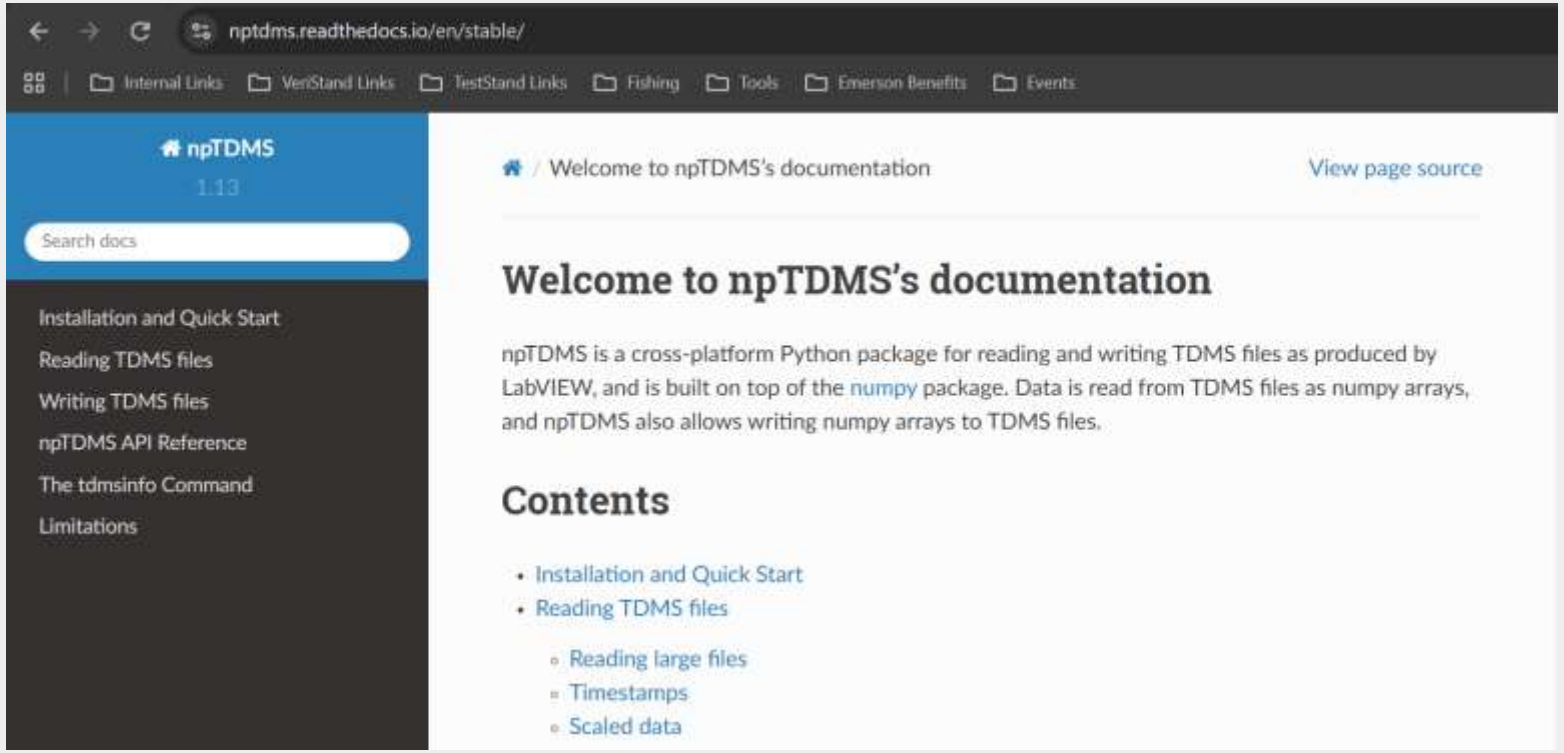
Result:

No Problem!!



TDMS Python API

https://nptdms.readthedocs.io/en/stable/



NI TDMS Python API

Prompt

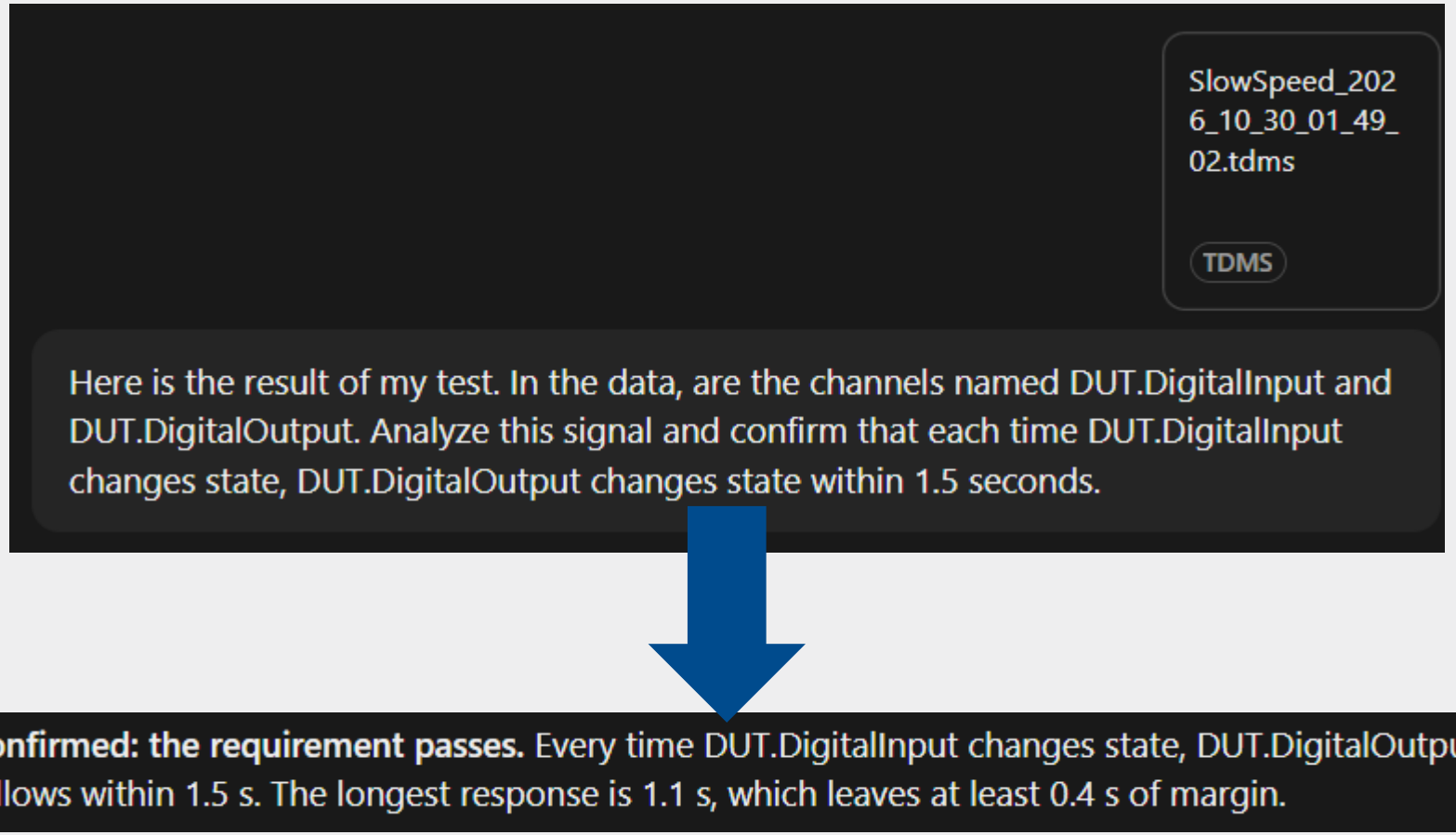
Here is my resulting TDMS data

Analyze DUT Channels

Compare against test requirement

Result:

No Problem!!

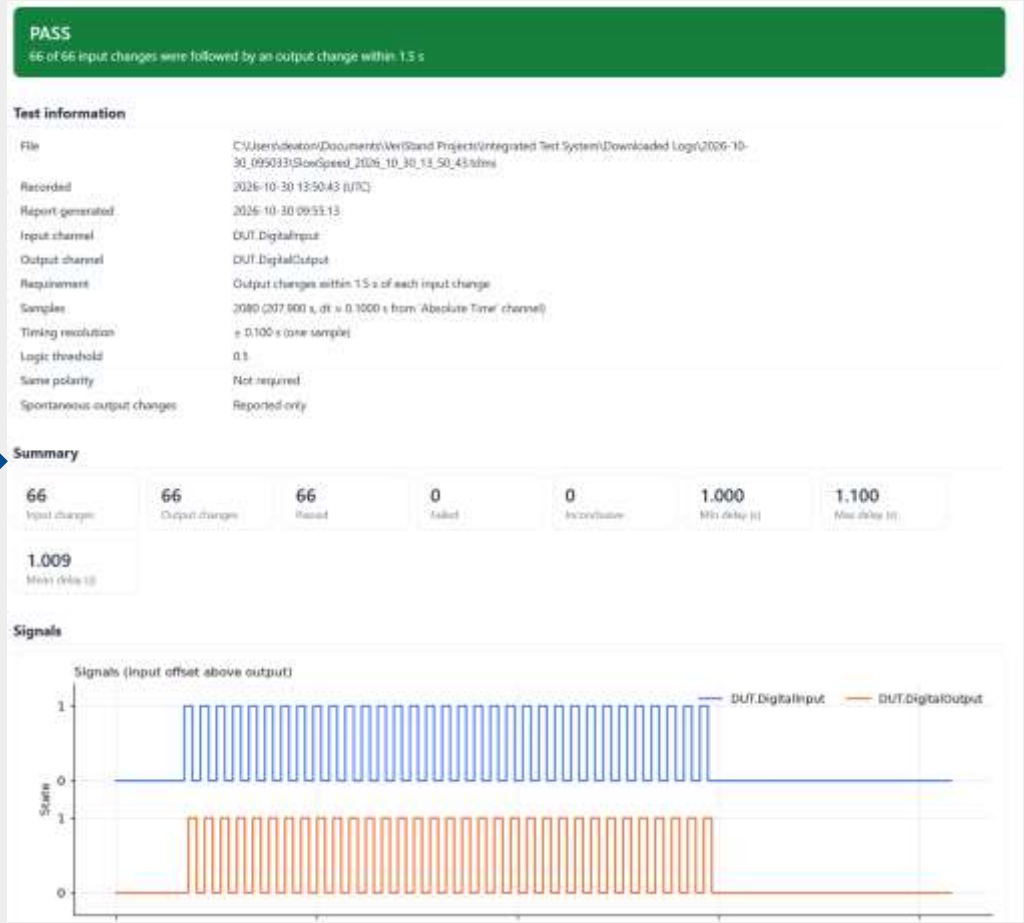


NI TDMS Python API

Prompt

Create Python Script with Report

Write a python script that does this analysis for me.



NI LabVIEW API

Prompt

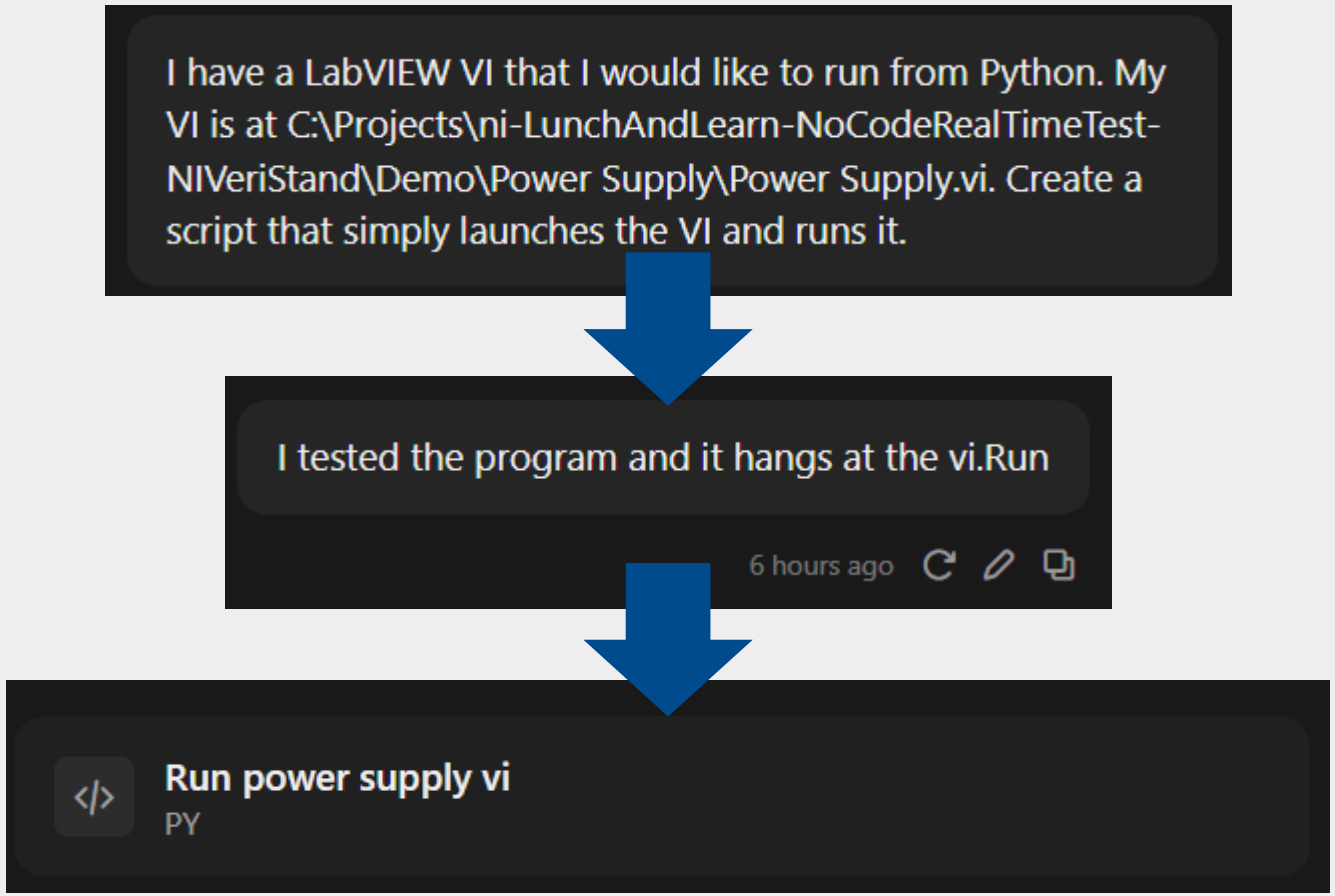
Start VI Before VeriStand

Result: Runs but hangs

Tell it to do it better…twice…

Result:

No Problem!!



Run the Demo!

What does it do?

Script:

- Opens LabVIEW
- Runs a LabVIEW VI
- Opens VeriStand
- Runs VeriStand Project (including user created plugins and models)
- Runs VeriStand sequence
- Downloads resulting data files
- Runs automated analysis on specific TDMS files
- Generates a report

Run the Demo!

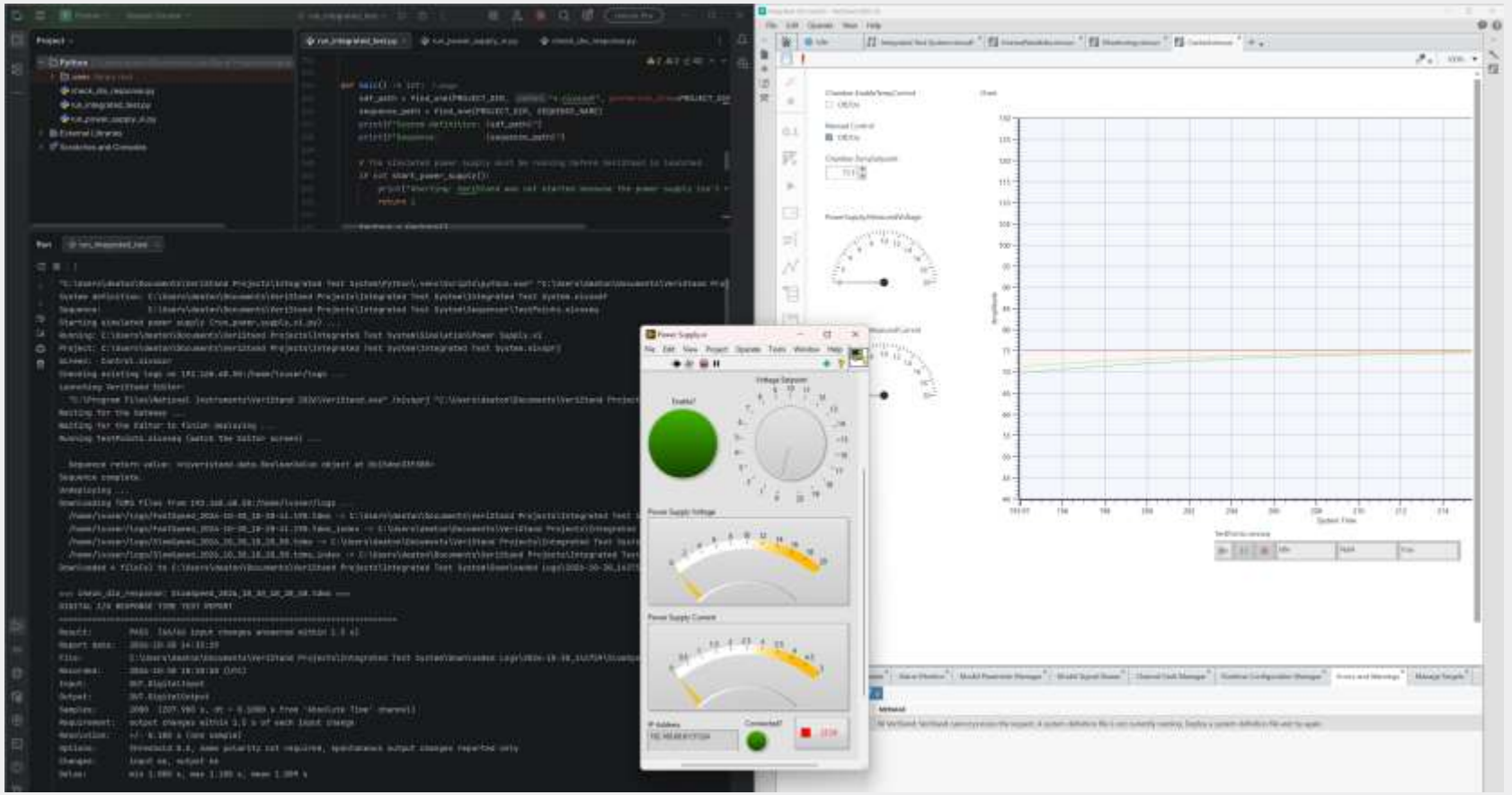
Open run_integrated_test.py

Review code

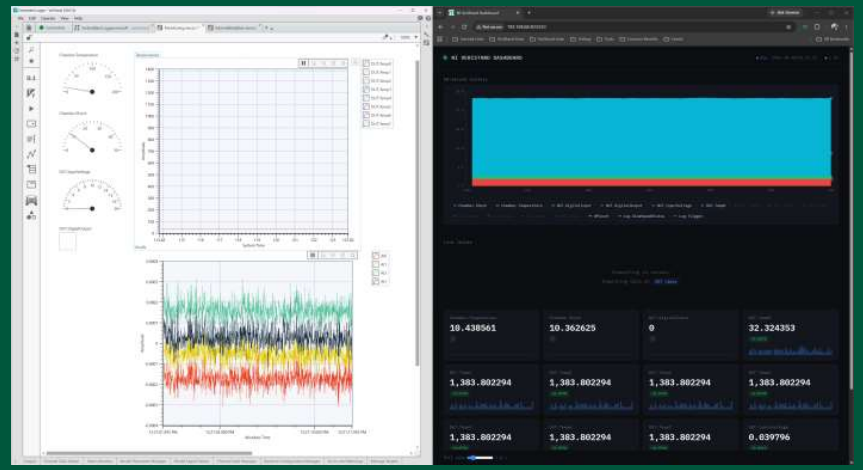
Run

Assumes:

- Venv setup
- \Python\PIP Requirements.txt installed



Demo: NI VeriStand as Integrated Test System Summary



Extend VeriStand from the inside (custom devices, models, tools)

Extend VeriStand from the outside (public APIs)

(Check Github Community for VeriStand extensions)

NI owns the core frameworks, APIs, and interfaces

You own the extension, not the architecture

NI Products work amazingly well with Vibe Coding…

Latest NI Investment in VeriStand

EMERSON

Getting Start Resources

Existing Resources

Read the help (link)

Get a support plan (link)

Watch the YouTube Series (link)

Take the online training (link)

Onsite training (talk to your account manager)

NEW Resource



Nigel AI

Nigel AI is your trusted test and measurement companion, assisting you in your daily tasks with expert information and guidance while keeping you in control of every engineering decision.

Explain Test Applications and Methodologies
Get instant information on test and measurement concepts and learn best practices to decrease your development time

Design and Build Your Test System
Get hardware recommendations for your application, compare specifications, and make informed decisions

Quickly Onboard to NI Software
Nigel accelerates onboarding with software guidance and provides examples demonstrating core concepts and functionality

Develop Faster
Nigel generates code and guides development in NI Software specific to your exact project to eliminate roadblocks



EMERSON

Lunch and Learn on Getting Started w/ Nigel

- Presentation: https://github.com/NISystemsEngineering/ni_LunchAndLearn-NIVeriStandNigelAssistantExamples
- Videos on everything!
- Build along!



EMERSON

Questions?

WE WANT TO HELP!!

- Set up follow-up meetings with account managers in the room



NI is now part of Emerson.